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FIRMWARE BUILD AND END-OF-LINE PROVISIONING

Document: FW-EEG-001 **Revision:** C **Date:** 2026-09-01 **Issued by:** TI One Voice research programme (one.witysk.org), Brussels, Belgium **Licence:** CC BY-SA 4.0 for this document; the firmware source is MIT **Governing documents:** DSN-EEG-003 Rev C, then RFQ-EEG-001 Rev E. Where this document and design.py disagree, design.py governs.

Revision note (Rev C): carries the layout findings – the carrier is now 150.0 x 130.0 mm and four layers – and **rules** the ring-buffer depth to a 6 MiB ring, 126 s at 1000 Hz, against the relaxed F-06 (ECO-EEG-025). That was a ruling, not a report of work done: when Rev C was issued `main.c` still asked for 12 MiB, which was FW-D13 and open. The source has since been changed and FW-D13 is closed – see the correction note below. It corrects the frame bandwidth to 50.7 kB/s, the end-of-line flashing route to the DevKitC-1's own UART port, the E-28 position, the eFuse phasing and the contact-light driver status. The findings of the second cross-document audit of 1 September 2026 that name this document – the layout-rules citation in section 3.1, the ring-buffer arithmetic in section 5.8, and the microSD file layout, which this document now owns in section 5.9 – are closed in this issue. The revision letter does not move: these are corrections within one release, not a new one.

Correction note, 2 September 2026. The firmware source was changed on 2 September and this document is re-synchronised against it here. An independent review found that Rev C described the *v1* source rather than the shipped one, and that matters more than ordinary staleness: a specification is the contract a builder follows, and as issued this document told them to re-introduce defects the source no longer had. What changed, and where this document changed with it:

- `firmware/main/drivers.h` is **new**. `main.c` had been calling seven functions that `drivers.c` defines with no declaration in scope. An implicit declaration is an error in C99 and later, and ESP-IDF compiles with `-Werror=implicit-function-declaration`, so the project could not have been built as shipped; worse, where a compiler accepts it the return type is assumed to be `int`, which would have truncated `sd_free_mb()`'s `uint32_t` and misread the pointer returns with no diagnostic at all. Section 1.2.
- `main.c` now includes `board_pins.h`, `esp_attr.h` and `drivers.h`. **FW-D12 and FW-D13 are closed in the source**, not merely ruled: the Rev A pin block is gone and `RING_BYTES` is 6 MiB at `main.c:101`.
- **FW-D14 is fixed**. `rx_poll()` strips the ten-byte frame header and checks the protocol version and the frame type before dispatch. It had been passing the whole frame, so `handle_command()` read header byte 0 – the protocol version – as the opcode; since `PROTO_VERSION` is 1 and `CMD_START_SESSION` is `0x01`, every command the browser tool sent started a recording session. Sections 1.1 and 6.1.
- The `CMD_ACK` payload is reshaped to section 6.2 exactly, in `ack_emit()`. Section 6.2 also records the one path that does **not** yet use it – provisioning, now FW-D20.

- `CMD_IDENTIFY` sets `CAP_CODEEC`, which nothing had ever set, so a working codec always read as absent; and it separates `CAP_ATECC` (the part is fitted and answering) from `CAP_PROVISIONED` (its configuration zone is locked), which had been conflated, so an unprovisioned board coming off the line reported no secure element at all. Section 6.3.
- Section 6.3's opcode table assigned `0x0F` to `FW_UPDATE_END`, colliding with the `CMD_IDENTIFY` that `main.c` and `TOOL-EEG-022` have used since 1 September, and had no row at all for `0x10 CMD_LOOPBACK`. The table now carries both commands and `FW_UPDATE_END` has moved to `0x18`.
- `webtest/tests/interop/` is **new**: it compiles the real `main.c` against small ESP-IDF stubs on a development host and drives it with the real browser-tool protocol module. Section 8. This was the first time any of this firmware had been compiled or executed at all, and it was not a build against ESP-IDF and not a run on hardware. **Superseded the same day** – see the build note immediately below.

Build note, 2 September 2026. The firmware was **built** later the same day, against a real ESP-IDF installation, and then **run under QEMU**. Both are new facts and both are narrower than they sound; the whole of this document is re-synchronised against them here, because a specification that still says “never compiled” after the compile is a document a reader has to second-guess.

- The build is **ESP-IDF v5.2.5**, target `esp32s3`, from `sdkconfig.defaults` plus `sdkconfig.phase1`. Section 2.1 pinned **v5.2.2** and is corrected. The four release images and a manifest of their SHA-256 are in `firmware/release/`; section 9 carries the figures.
- Getting there took five corrections to the **project**, not to the protocol, and each is recorded in the file it was made in: `esp_driver_i2c` does not exist at IDF 5.2 and the component list would not configure; `CONFIG_BOOTLOADER_APP_ANTI_ROLLBACK=y` had been appended to `sdkconfig.phase1` below the deliberate `=n`, where a later duplicate silently wins, and ESP-IDF refuses anti-rollback on a table carrying a factory partition, so the build stopped; `CONFIG_TINYUSB_VENDOR_ENABLED` is not an option name that exists in `esp_tinyusb 1.4.x`, so the vendor interface this device is built around was silently absent – the real key is `CONFIG_TINYUSB_VENDOR_COUNT`; and three descriptor callbacks were defined here **and** in `esp_tinyusb`, which is FW-D02, and the link failed until the three were deleted and the component left to own them; and the five `sdkconfig` lines this document had listed as “must be added before the first build” were still not added. Sections 2.2 and 2.4.
- **FW-D02, FW-D03, FW-D11 and FW-D16 are closed, and FW-D20 is closed.** The `_Static_assert` on the MS OS 2.0 descriptor length has now been evaluated against resolved macros and held; the bicolour contact-light phase driver is written, so **E-27 is met in the source** and the FW-D16 row is retired; and `handle_provision()` now answers through `ack_emit()`, so the provisioning family is the section 6.2 acknowledgement. Sections 1.3, 3.2 and 6.2.
- Two opcodes were added: **`0x4A CMD_READ_CALIBRATION` and `0x4B CMD_ATECC_WRITE_CONFIG`**. Section 6.3's table did not carry them and now does.
- Section 7.3's table said it was `provision.py`'s step list. It was not, and the divergence was not cosmetic: the script writes and locks the ATECC configuration

zone **before** GenKey, because a factory-fresh part will not generate a key into an unconfigured slot. That section is rewritten from the script.

- The QEMU run is in `firmware/release/qemu_boot.log`. **It is an emulator, not a unit**, and section 8.2 states exactly what it does and does not prove.
- The build produced two memory figures and one of them is a finding. The image is **405,245 bytes** in a 3 MB slot, which is ample. **Static IRAM is 16,383 of 16,384 bytes used – one byte free**, which is a cliff and not a margin, and taking two unused ISRs out of IRAM did not move it by a byte. Section 9 records both and section 10 item 17 carries the open one.
- Every line count and line-number citation in sections 1.1, 1.2 and 1.3 was stale by up to 130 lines and is re-counted; the interop harness reports **57 checks**, not the 32 section 8.1 quoted.

The revision letter does not move: nothing here is a new release. Two things a reader tracks **do** change, and they are called out rather than buried. The ESP-IDF pin moves from **v5.2.2 to v5.2.5**, which is the version that produced the shipped images, and anyone reproducing the build must use it. And **E-27 moves from not met to met in the source**, because the phase driver it was waiting on is written – met in the source is not signed off, and TST-EEG-004 T11 still needs a unit and a colorimeter. No pin, no part and no quotable quantity changes.

Why this document exists

Package v1 shipped one file, `firmware/main.c`, and nothing around it. There was no project, no configuration, no partition table, no pin map that matched the board, no provisioning script and no host tool – yet RFQ-EEG-001 section 1.1 asks a manufacturer to price “firmware loading and provisioning at the end of the line (we supply the firmware image and the provisioning script)”, and TST-EEG-004 makes running that script a test step. This document specifies everything that was missing, states exactly what the source does and does not do today, and defines the byte-level contract that the firmware, the manufacturer’s test station and the programme’s browser session runner must all agree on.

Corrected 2026-09-02. This paragraph read “**Nothing in this package has been built, and no safety engineer has reviewed this design.** Nothing here has been compiled or run on hardware.” Half of that is no longer true and the half that is withdrawn is withdrawn here rather than deleted: the firmware **was** compiled, against ESP-IDF v5.2.5, on 2 September 2026, and the images are in `firmware/release/`. What still stands, unchanged and in full:

No safety engineer has reviewed this design. No board has been fabricated and no unit exists. The firmware has run only under QEMU, whose esp32s3 machine emulates none of this instrument’s peripherals (section 8.2). Every number marked *calculated* is still calculated, and every register value, daisy order and SPI timing in this document is still an assumption that no silicon has answered.

1. What exists and what is stubbed

The reference source is no longer one file. It is `firmware/main/main.c` (832 lines), `firmware/main/drivers.c` (580 lines) and `firmware/main/drivers.h` (77 lines), over the generated `firmware/main/board_pins.h` (74 lines); counted with `wc -l` on the shipped files at 2 September 2026. *Corrected 2026-09-02: this paragraph said 705,*

476 and 69, which were the counts before the phase driver, the two new provisioning opcodes and the acknowledgement rework landed. `main.c` was 362 lines in v1, and the growth is the provisioning handler of section 7, the acknowledgement of section 6.2, the contact-light phase driver of section 3.2, the two connectivity commands of TOOL-EEG-022 and the receive-path fix.

It builds. *Corrected 2026-09-02: this paragraph used to say the source had “never been built against a real ESP-IDF installation”.* It has been, once, on 2 September 2026: ESP-IDF **v5.2.5**, target `esp32s3`, `sdkconfig.defaults + sdkconfig.phase1`, clean at ESP-IDF's default `-Wall -Werror=all`. The four images and their SHA-256 are in `firmware/release/` and section 9 carries the figures. That closes the *build* half of the FW-D01 exit criterion and it retires nothing else.

Read the rest of the claim narrowly, because three different things have now happened to this source and they prove three different amounts:

What ran	What it proves	What it does not
<code>sh webtest/tests/interop/run.sh – main.c</code> compiled on a development host against small ESP-IDF stubs and driven by the real browser-tool protocol module: 57 checks, all passing	the firmware and the host tool agree about framing, the CRC, the header, the opcodes, the acknowledgement layout, the S-01 interlock and the 0x4A/0x4B split	nothing about ESP-IDF, TinyUSB, the descriptors or any peripheral: the stubs expand the descriptor macros to placeholders
<code>idf.py</code> build against ESP-IDF v5.2.5	every ESP-IDF header, Kconfig option and TinyUSB macro this file names resolves; the <code>_Static_assert</code> on the 10-byte frame header and the one on the MS OS 2.0 descriptor length both held; the image links and fits	that any of it is <i>right</i> . A register value that assembles is not a register value the ADS1299 accepts
<code>qemu-system-xtensa -M esp32s3</code> – one full boot, <code>firmware/release/qemu_boot.log</code>	the image boots, the bootloader reads this partition table, the app loads from the factory slot, <code>app_main()</code> runs, and the microSD, codec and ring-buffer paths behave as written	every peripheral. QEMU's <code>esp32s3</code> has no octal PSRAM, no microSD, no ES8388 and no ADS1299. Section 8.2

It has still **never run on hardware**, so the daisy order, the SPI timing, every register value and the whole analogue front end remain assumptions. The two C++ range-based for loops that made the v1 file a syntax error in C are gone (FW-D01).

The following inventory is what a reader will find in the file today.

1.1 Implemented in full

Line numbers are main.c at 2 September 2026, **re-counted after the build**. *Corrected 2026-09-02: every line number in this table was stale, by up to 130 lines at the foot of the file.*

Function	Lines	What it does	Confidence
crc32()	115-119	IEEE 802.3 reflected CRC-32, polynomial 0xEDB88320, init 0xFFFFFFFF, final inversion. Matches Python binascii.crc32	High – byte-checkable offline, and checked against the host implementation by the interop harness
cobs_encode()	121-128	Standard COBS with the trailing 0x00 delimiter written by the encoder	High – decoded by the browser tool in the interop harness
frame_hdr_put())	147-159	Serialises the ten-byte header field by field, with a <code>_Static_assert</code> on the length (FW-D19)	High – the harness parses the result with the host decoder, and the assert held in the ESP-IDF build
frame_emit()	167-184	Builds header, appends payload, appends CRC, COBS-encodes, writes to CDC, vendor, ring buffer and SD	Structurally complete, not thread-safe (see FW-D04)
ads_cmd(), ads_wreg_all() , ads_set_rate()	187-198	Single-byte opcodes and daisy-chain register writes	Register values plausible, never verified against silicon
ads_init()	199-224	SPI bus, reset sequence, CONFIG1/2/3, per-channel gain, lead-off, MISC1, RDATA_C	Compiles and links; never run against a converter, in QEMU or anywhere else
drdy_isr()	237-247	Increments <code>g_sample_index</code> , latches the envelope-comparator level and records its rising edge (FW-D08), notifies the sample task. The only place the counter moves (E-19)	Correct in principle
aux_bits()	248-257	Builds six of the sixteen auxiliary bits	Incomplete, see section 5.3

Function	Lines	What it does	Confidence
sample_task()	258-282	54-byte daisy read, de-interleaves the two devices, packs 50-byte sample records, emits one frame every 20 ms	Never run; daisy order is an assumption
lights_write()	285-288	Shifts eight bits into the 74HC595 and latches	Never run
lights_phase() , lights_task()	325-360	The bicolour phase driver (E-27). Phase A lights the wanted sites green with LED_V high, phase B lights them red with LED_V low, a site in both is amber; the masks come from the converter's positive-side lead-off comparator read at two thresholds – trips neither is green, the sensitive one only is amber, both is red (FW-D17). Dark while recording, and LED_V is returned to an input rather than driven low, which is dark by construction	Written, on 2 September 2026 – FW-D16 is closed and E-27 is met in the source. Never run. The alternation quantises to the FreeRTOS tick: nominal LIGHT_PHASE_HZ 240, actual about 250 Hz – section 3.2
ack_emit()	430-439	The section 6.2 acknowledgement: opcode echoed, reserved zero, status, result length, result	High – decoded by the browser tool's own parseAck() in the harness
handle_provision() ()	441-561	The thirteen 0x40-0x4B and 0x4F provisioning opcodes of section 7, including 0x4A CMD_READ_CALIBRATION and 0x4B CMD_ATECC_WRITE_CONFIG	Never run against an ATECC608B. Its acknowledgement is the section 6.2 one as of 2 September – FW-D20 closed
handle_command() ()	563-658	Dispatches eleven general opcodes – 0x01-0x05, 0x08-0x0B, 0x0F, 0x10 – and routes 0x40-0x4F to handle_provision()	Five declared opcodes still fall to the unknown-opcode arm: FW-D10

Function	Lines	What it does	Confidence
timing_selftest()	664-670	40 tone bursts, insertion sort, median and p95	The two functions it calls now exist in <code>drivers.c</code> ; neither has driven a codec
USB descriptor tables, tud_descriptor_bos_cb()	673-722	Device, configuration, string and BOS descriptors. <i>Corrected 2026-09-02:</i> there are not four <code>tud_descriptor*_cb</code> callbacks here any more. <code>esp_tinyusb</code> owns the device, configuration and string callbacks and is handed those three as data through <code>tinyusb_config_t</code> ; defining them here as well is FW-D02 and it failed the link with three “multiple definition” errors. Only <code>tud_descriptor_bos_cb()</code> stays, because <code>esp_tinyusb 1.4.x</code> defines none and <code>tinyusb_config_t</code> has no field for one	Macro names now resolved against the pinned <code>esp_tinyusb</code> by the real build, and <code>MS_OS_20_DESC_LEN</code> asserted. Never enumerated by a host
tud_vendor_control_xfer_cb()	723-734	Serves the WebUSB URL and the MS OS 2.0 descriptor	Never run
rx_poll()	736-763	One pass of the receive loop: COBS-decode, check the CRC, skip the ten-byte header and check the version and the frame type , then dispatch (FW-D14). Factored out of <code>rx_task()</code> so a host harness can drive the real path rather than a copy of it	Exercised end to end by the interop harness
rx_task()	765-770	Calls <code>rx_poll()</code> , and yields for 2 ms when it read nothing	Never run

Function	Lines	What it does	Confidence
status_task()	773-778	One STATUS frame per second	Payload is still 12 bytes carrying four of the eight items F-09 names, with link type at offset 8 where section 5.4 puts uptime (FW-D09, open)
app_main()	780-825	GPIO setup, drv_init_all(), serial read, ring buffer with an explicit abort if PSRAM will not give it, TinyUSB install, ADS init, four tasks	Runs to completion in QEMU as far as the ring-buffer allocation, which QEMU cannot satisfy and which aborts with the named diagnostic FW-D13 added. Never run on a target

1.2 The drivers are written now, and none of them has run

At v2.1 these six symbols were declared extern **inside function bodies** and defined nowhere in the package. They are now declared in firmware/main/drivers.h and defined in firmware/main/drivers.c, which is in SRCS (section 2.2). Writing a driver and proving one are different claims and only the first has happened: not one of these has addressed a real peripheral, so every I2C address, register write and timeout below is still an assumption, and the test steps in the last column still gate on hardware.

Line numbers corrected 2026-09-02.

Symbol	Defined at	Driver	Called from	Blocks
void sd_append(const uint8_t *, size_t)	drivers.c:163	SDMMC, one-bit	frame_emit(), every frame	T14
uint32_t sd_free_mb(void)	drivers.c:175	SDMMC	status_task(), CMD_IDENTIFY	T14, F-09
void atecc_serial_into(char *, size_t)	drivers.c:288	ATECC608B I2C	handle_provision() at 0x43	T5b, T6, F-04
uint8_t battery_percent(void)	drivers.c:107	MAX17048 I2C	status_task()	F-09, E-22
void codec_play_tone_at(uint32_t)	drivers.c:234	ES8388 I2S	timing_selftest()	T12, T13, T17

Symbol	Defined at	Driver	Called from	Blocks
int envelope_onset _after(uint32_ t, int)	drivers.c:2 50	signal	timing_selfte st()	T12, T13

drivers.c also defines unit_serial_into(), the drv_atecc_* and drv_nvs_* groups section 7 needs – including drv_atecc_write_config() at drivers.c:431 and drv_nvs_get_blob() at drivers.c:552, both added on 2 September for the two new provisioning opcodes – and drv_codec_ready() and drv_atecc_present(), which are what CMD_IDENTIFY reports as CAP_CODEC and CAP_ATECC.

The QEMU run exercised three of these paths and answered a question about all of them. drv_sd_init() timed out, drv_codec_init() timed out on ES8388 register 0x00, and both logged the failure and continued rather than aborting, which is what they were written to do. That is a bring-up behaviour test and nothing more: QEMU emulates neither part, so a timeout is the only answer it could ever have given, and no I2C address, register write or timeout below has yet been answered by a part.

drivers.h exists because it had to. main.c called seven of these functions with no declaration in scope, which is an error in C99 and later and which ESP-IDF rejects at -Werror=implicit-function-declaration; and where a compiler does accept an implicit declaration it assumes a return type of int, so sd_free_mb()'s uint32_t and the pointer returns would have been truncated or misread at runtime with no diagnostic. Include it from both files, so the compiler checks the definitions against the prototypes the callers use.

A seventh item is worse than a stub: the **block-signing task is not declared at all**. The closing comment says SIGNATURE frames are “emitted as FT_SIGNATURE by the sd/signing task (not shown)”. Nothing emits them, so T16 – the only place the study’s integrity mechanism is ever tested – has no input. That is unchanged at 2 September: drivers.c contains no SHA-256, no chained digest, no signing call and no such task, and F-08 is therefore unimplemented rather than merely untested.

1.3 The bring-up defect register

These are the defects found by reading the file, and – from 2 September 2026 – by building it. Each is a work item with an acceptance test. Anyone quoting firmware bring-up should quote against this table, not against “complete the stubs”. Twenty entries; **fourteen are fixed or closed in the source and six are open**. *Corrected 2026-09-02: this line read “nine are fixed and eleven are open”. FW-D02, FW-D03, FW-D11, FW-D16 and FW-D20 have closed since, four of them because the first real build forced them into the open.*

A row marked FIXED means the source no longer has the defect. It does not mean the acceptance test has been run: the acceptance column is the gate, and for most of these rows that gate is a unit on a bench, which does not exist. **The build gate, however, is now real** – idf.py build against ESP-IDF v5.2.5 completes clean at ESP-IDF’s default -Wall -Werror=all – so where a row’s acceptance was “build once”, that row is genuinely retired and the column says so.

The six open rows are **FW-D04, FW-D05, FW-D06, FW-D07, FW-D09 and FW-D10**. Five of the six are frame-integrity or protocol-completeness items and none of them is

visible to a compiler; the sixth, FW-D09, is a wire-format shortfall a host decodes wrongly in silence.

ID	Defect	Violates	Fix	Acceptance
FW-D01	C++ range-based for in app_main(), twice	compiles as C	FIXED: explicit arrays and an index loop, main.c:783-793	MET 2026-09-02. idf.py build against ESP-IDF v5.2.5 completes clean at ESP-IDF's default -Wall -Werror=all. A full -Wextra -Werror build has not been demonstrated and is not what ESP-IDF sets

ID	Defect	Violates	Fix	Acceptance
FW-D02	Descriptor callbacks defined and the same descriptors passed to tinyusb_driver_install()	link error	FIXED 2026-09-02, and the predicted fix was the wrong one. This row used to say “set CONFIG_TINYUSB_DESCRIPTOR_CUSTOM=y and keep the callbacks”. Doing both is exactly the defect: with CONFIG_TINYUSB_DESCRIPTOR_CUSTOM=y the descriptors are supplied as data through tinyusb_config_t and esp_tinyusb owns the callbacks. The link failed with three “multiple definition” errors against descriptors_control.c until tud_descriptor_device_cb(), tud_descriptor_configuration_cb() and tud_descriptor_string_cb() were deleted from main.c. tud_descriptor_bos_cb() stays, because the component defines none	Build gate MET (the image links). T5 for enumeration

ID	Defect	Violates	Fix	Acceptance
FW-D03	The TinyUSB vendor class is not enabled, so tud_vendor_* has no backing interface	F-01	FIXED 2026-09-02, and the option name in this row was wrong. CONFIG_TINYUSB_VENDOR_ENABLED does not exist in esp_tinyusb 1.4.x, and ESP-IDF does not fail on an unrecognised Kconfig key – it warns and carries on. So the setting looked present, CFG_TUD_VENDOR stayed 0, tusb.h declared none of the tud_vendor_* functions, and the WebUSB half of F-01 would simply not have existed. It surfaced as “implicit declaration of function tud_vendor_mounted” the first time the file met a real compiler. The real key is CONFIG_TINYUSB_VENDOR_COUNT=1 , now in both sdkconfig.defaults and sdkconfig.phase1	Build gate MET. T5 for enumeration
FW-D04	One static txbuf/cobsbuf pair written by four callers at priorities 23, 10, 5 and 4 with no mutex	frame integrity	a recursive mutex around frame_emit(), or per-task encode buffers	T14, 30 min, zero bad CRC
FW-D05	The overflow path drops the oldest frames and emits no GAP	F-07 “silent loss is not permitted”	emit FT_GAP per section 5.5	T15

ID	Defect	Violates	Fix	Acceptance
FW-D06	CMD_RETRANSMIT ignores its range and drains the ring destructively	F-06	index the ring by sequence, copy without consuming	T15 complete recovery
FW-D07	g_seq++ from three tasks, non-atomic	frame ordering	increment under the FW-D04 mutex	T14
FW-D08	The comparator aux bit is sampled in the task, not latched in the DRDY ISR	E-12 sub-sample onset	FIXED: drdy_isr() latches the level and records the rising edge in g_onset_sample, and aux_bits() reports the latched value (main.c:235-255)	T12b
FW-D09	STATUS carries battery, lead-off, SD free and link type only, in a 12-byte payload, and link type sits at offset 8 where section 5.4 puts uptime, so a host parsing per section 5.4 decodes it as part of uptime	F-09 (8 items)	the 24-byte, 10-field payload of section 5.4, which includes the ring depth T15 grades against (section 5.8)	T14, T21, T15

ID	Defect	Violates	Fix	Acceptance
FW-D10	Open, and unchanged by the build. CMD_SET_GAIN, CMD_IMPEDANCE, CMD_PLAY_AT, CMD_FW_UPDATE_BEGIN and CMD_PROVISION are still declared in the enum (main.c:363-369) with no case arm, so all five reach the default: arm. The status they now return is 0x01, unknown opcode – not 0xFF any more, and not the 0x0B not-implemented that section 6.2 defines and that a host needs in order to tell a missing feature from a typo. Of the five, only CMD_PROVISION is answered elsewhere: provisioning has moved to the 0x40-0x4F family, which handle_provision() implements in full, so 0x0E should be deleted from the enum rather than written	F-10	section 6; and return 0x0B, not 0x01, where the honest answer is “not implemented in this build”	T10, T11, T12, T17

ID	Defect	Violates	Fix	Acceptance
FW-D11	<code>_Static_assert(sizeof(ms_os_20_desc) == 0xB2)</code> never checked against resolved macros	F-02	CLOSED 2026-09-02: the assert (<code>main.c:705</code>) was evaluated in the ESP-IDF v5.2.5 build against the pinned <code>esp_tinyusb</code> and held, as did the ten-byte header assert at <code>main.c:160</code> . Every TinyUSB macro this file names now resolves. What that does not settle is whether Windows accepts the descriptor set	T5 on Windows 11
FW-D12	<code>main.c</code> carried the Rev A pin defines: <code>PIN_SR_DATA 35</code> , <code>PIN_SR_CLK 36</code> , <code>PIN_SR_LATCH 37</code> , which are the octal PSRAM bus, so a unit flashed with it would have torn down the memory the ring buffer lives in	ECO-EEG-009	FIXED: the block is deleted and <code>main.c</code> includes the generated <code>board_pins.h</code> , which comes from <code>design.py</code> , so the firmware and the board cannot disagree	T11

ID	Defect	Violates	Fix	Acceptance
FW-D13	RING_BYTES was 12 MiB; the -N16R8 carries 8 MiB of PSRAM in total, so the allocation returned NULL and the first DATA frame asserted inside xRingbufferSend	F-06	FIXED: RING_BYTES is 6*1024*1024 at main.c:101, and app_main() now aborts with a named diagnostic if PSRAM will not give it. F-06 relaxed by ECO-EEG-025 – section 5.8. The diagnostic has been seen to work: it is the last line of firmware/release/qemu_boot.log, because QEMU has no PSRAM at all to allocate from	T15
FW-D14	rx_task() treated the first decoded byte as the opcode; the host tool sends a full frame header first. The consequence was worse than non-interoperation: byte 0 is the protocol version, PROTO_VERSION is 1 and CMD_START_SESSION is 0x01, so every command from the browser tool started a recording session, and the one command that appeared to work did so by that coincidence	interoperability	FIXED on 2 September 2026: rx_poll() dispatches from out + FRAME_HDR_BYTES and refuses any frame whose version is not PROTO_VERSION or whose type is not FT_CMD (main.c:609-636)	32 of 32 in webtest/tests/interop; a provisioning dry run against hardware is still outstanding
FW-D15	<i>Closed at Rev C.</i> The header said “written against RFQ-EEG-001 Rev B” while the package shipped Rev E	traceability	done: this document is headed RFQ-EEG-001 Rev E and section 5 is written against it	closed

ID	Defect	Violates	Fix	Acceptance
FW-D16	The bicolour contact-light phase driver did not exist. <code>lights_write()</code> and <code>lights_task()</code> drove the 74HC595 on and off only; nothing alternated LED_V against the shift-register outputs, so no site could show red or amber	E-27, RFQ E-27's amber state	FIXED 2026-09-02: <code>lights_phase()</code> and <code>lights_task()</code> (<code>main.c:325-360</code>) drive LED_V (GPIO48) and Q0-Q7 in antiphase, and the three colours come from the converter's positive-side lead-off comparator read at two thresholds – trips neither is green, the sensitive threshold only is amber, both is red (as corrected by FW-D17, below; as first written this used <code>LOFF_STATP</code> & <code>LOFF_STATN</code> , and red was unreachable). E-27 is met in the source. The alternation quantises to the 1 kHz FreeRTOS tick and runs at about 250 Hz , not the nominal 240; the requirement is “above 100 Hz” and 250 Hz meets it. Section 3.2	T11. The driver is written, so T11 is no longer blocked by its absence; it still needs a unit and a colorimeter, and it has never been run
FW-D17	CMD_START_SESSION armed a session without ever reading PIN_VBUS_DET. Only the CHG_CE half of S-01 existed, so a unit tethered to a mains-powered charger would record	S-01	FIXED at Rev C: the handler refuses first and returns ack status 0x05, per section 6.1	T21, and the interlock half of T3

ID	Defect	Violates	Fix	Acceptance
FW-D18	CMD_SET_RATE issued WREG CONFIG1 while the converter was in RDATA; the ADS1299 ignores register writes in that mode, so the rate silently did not change	E-02	FIXED: START low, SDATA, write CONFIG1, RDATA, START high (main.c:512-523)	T2 at all three rates
FW-D19	The frame header was memcpd onto the wire with sizeof(frame_hdr_t), which is 12, not 10. The struct is tail-padded to its 4-byte alignment, so two padding bytes sat between the header and the payload and every host parser written to section 5.1 – including this package's own verify_stream.py – misparsed every frame by two bytes	section 5.1	FIXED at Rev C: the header is serialised field by field by frame_hdr_put(), with a _Static_assert on the 10-byte length. A wire format is a contract with other people's software and must never be a struct copy	T5b, and TOOL-EEG-022 step D4

ID	Defect	Violates	Fix	Acceptance
FW-D20	The provisioning acknowledgements were not the section 6.2 acknowledgement. <code>handle_provision()</code> built {opcode, status, result...} and called <code>frame_emit()</code> directly, while <code>ack_emit()</code> and every other reply send {opcode, reserved 0, status, result length, result...}. <code>provision.py</code> reads the status at absolute frame offset 12, which is payload offset 2, so it read the first <i>result</i> byte as the status: on 0x48 <code>READ_PROVISION_STATE</code> that byte is the config-zone lock flag, so an unlocked unit reported 0 and looked like a success – the one command whose whole job is to say whether provisioning happened answered “fine” for a blank part	section 6.2	FIXED 2026-09-02: the whole 0x40-0x4F family returns through <code>ack_emit()</code> (<code>main.c:561</code>), including the not-in-provisioning-mode refusal. Status codes are unchanged; only the envelope moved	T6. The interop harness now covers the shape: three of its 57 checks read a provisioning status at the section 6.2 offset, and one of them is a 0x02 bad-length refusal that would have decoded as something else under the old shape. A run against a real ATECC608B is still outstanding

Exit criterion for the whole backlog: a clean build at `-Werror`, enumeration on Windows 11, macOS and Ubuntu (T5), and T13 reporting median ≤ 1 and p95 ≤ 2 samples. *Corrected 2026-09-02: the first of those three is now met* – the ESP-IDF v5.2.5 build is clean at `-Wall -Werror=all`. **The other two are not, and they are the ones that need a unit.** The host harness of section 8 is not that build and retires nothing; it is a guard against the firmware and the host tool drifting apart again. The QEMU run of section 8.2 retires nothing either: a boot on an emulator with none of this instrument’s peripherals cannot grade a single row above.

2. Build environment

2.1 Toolchain

Corrected 2026-09-02. This table pinned **v5.2.2**. The firmware was built with **v5.2.5** – the released images in `firmware/release/` and their `manifest.json` say so – and a document that pins a version nobody used is a document a second builder will not reproduce. The pin moves to the version that produced the shipped images. **v5.2.2** remains inside the `>=5.2, <5.4` range `main/idf_component.yml` permits, so nothing about the component resolution changes.

Item	Value	Why it is pinned
ESP-IDF	v5.2.5 , tag <code>v5.2.5</code> ; <code>main/idf_component.yml</code> permits <code>>=5.2, <5.4</code>	The TinyUSB descriptor macros moved between releases (FW-D11), and this is the version the shipped images were built with
Managed component	<code>espressif/esp_tinyusb</code> <code>~1.4.2</code> , which resolved to 1.4.5 in this build	Same reason. <code>firmware/dependencies.lock</code> records what actually resolved and is part of the release
Target	<code>esp32s3</code>	E-18, not substitutable
Host	Ubuntu 22.04 or Windows 11, Python 3.11	The release container is Linux
Container	<code>espressif/idf:v5.2.5</code> , pinned by image digest	Reproducible build, section 9

The build is not yet reproducible in the sense section 9 asks for. The shipped images were built on a developer machine, not in the pinned container, and the boot log records the application version as `e91f9d58-dirty` – a working tree with uncommitted changes. Reproducing the SHA-256 list needs a clean tree and the container, and that has not been done.

An unpinned build is a silent difference between units, and section 9 of RFQ-EEG-001 exists because device identity must not become a confound in the study.

2.2 Repository layout

<code>firmware/</code>	
<code> CMakeLists.txt</code>	<code>project eeg_field_kit, VERSION 0.2.0</code>
<code> sdkconfig.defaults</code>	the fleet configuration (section 2.4)
<code> sdkconfig.phase1</code>	the Phase 1 overlay: no secure boot, no
<code>eFuses (section 2.4)</code>	
<code> sdkconfig.qemu</code>	the emulator overlay -- NOT a configuration
<code>for any unit (8.2)</code>	
<code> partitions.csv</code>	the flash layout (section 2.5)
<code> dependencies.lock</code>	what the component manager actually
<code>resolved</code>	
<code> README.md</code>	status and the three commands
<code> main/</code>	
<code> CMakeLists.txt</code>	<code>idf_component_register(...)</code>

idf_component.yml	pinned dependencies
board_pins.h	GENERATED from tools/design.py (section 3)
drivers.h	the peripheral API main.c calls (section 1.2)
drivers.c	the implementations of it (section 1.2)
main.c	the reference source
tools/	
provision.py	end-of-line provisioning (section 7)
verify_stream.py	host verification (section 8)
provision_selftest.py	offline checks on provision.py, including
opcode uniqueness	
atecc608b_config.py	generates the ATECC608B configuration-zone
template	
release/	the built images, their manifest, the size
report and the	
	QEMU boot log (sections 8.2 and 9). ADDED

2026-09-02

build/ is a build directory, not a release artefact, and is not part of the package.

The firmware also has a test that is not under firmware/: webtest/tests/interop/ compiles main.c against ESP-IDF stubs and drives it with the browser tool's protocol module. It lives next to the tool it proves agreement with, and section 8 describes it.

tools/provision.py and tools/verify_stream.py are the source-tree paths and are the names this package uses. Other documents in the release call the same two programs by other names; that divergence is recorded as an open item in section 10 and is not resolved here.

main/CMakeLists.txt as shipped:

```
idf_component_register(
  SRCS "main.c" "drivers.c"
  INCLUDE_DIRS "."
  # `driver` and NOT `esp_driver_i2c`. The I2C driver was split into
its own component
  # at ESP-IDF 5.3; at 5.2, which idf_component.yml also allows,
`esp_driver_i2c` does
  # not exist and the build stops at configure with "component could
not be found".
  # `driver` provides I2C on both, so it is the requirement that
spans the pinned range.
  REQUIRES driver esp_timer esp_psram sdmmc fatfs esp_partition
  app_update
           nvs_flash
  PRIV_REQUIRES esp_tinyusb
)
```

Corrected 2026-09-02. This block used to list esp_driver_i2c in REQUIRES. **That component does not exist at ESP-IDF 5.2** – the I2C driver was split out into its own component at 5.3 – and idf_component.yml permits $\geq 5.2, < 5.4$, so the build stopped at configure with “component could not be found”. driver provides I2C on both, so it is the requirement that spans the pinned range, and the line is now what is in the file.

The drivers came in as one drivers.c rather than the six files this section used to predict, and nvs_flash came in with them. What is still to be added is signing.c – the

block-signing task of section 1.2, which nothing implements – and an I2S component with it if the tone generator moves out of `drivers.c`.

2.3 Commands, and how a unit is put into download mode

```
. $IDF_PATH/export.sh                # ESP-IDF v5.2.5
idf.py set-target esp32s3
idf.py build                          # -> build/eeg_field_kit.bin
idf.py size                           # app size must stay under 3 MB
(section 2.5)
```

These are the commands that produced firmware/release/ on 2 September 2026. The Phase 1 build – which is what was actually run, and what the two prototypes take – adds the overlay:

```
idf.py -DSDKCONFIG_DEFAULTS="sdkconfig.defaults;sdkconfig.phase1" build
```

`idf.py size` reported a **405,245-byte** image against a 3 MB slot, and static IRAM at **16,383 of 16,384 bytes used**. The second of those is an open item, not a pass; section 9 gives both figures and section 10 item 17 carries the IRAM one.

Bench and end-of-line flashing both go through the **DevKitC-1's own UART USB-C port**:

```
idf.py -p /dev/ttyUSB0 -b 921600 flash monitor
```

That port is the only flashing route, and it is the route the production line uses.

The DevKitC-1 carries the auto-reset circuit on the module itself – the bridge's DTR and RTS lines drive EN and IO0 – so `esptool.py` puts the ESP32-S3 into download mode with no fixture, no relays and no operator action. The module is reachable with the pod lid off through the 31 x 61 mm opening in the MP-01 module plate, so the carrier does not have to be removed to flash a unit.

The carrier's J26 header is console and recovery only and cannot enter download mode. J26 carries 3V3, DGND, UART0 TX, UART0 RX, EN and one spare way netted NC_GPIO0. GPIO0 is committed to LED_SR_LATCH at J7 position 14 (ECO-EEG-009) and does not reach J26, so no relay sequence on J26 can assert IO0. Any procedure that describes flashing through J26 with a GPIO0/EN relay sequence is wrong. Console output is UART0 on GPIO43/44, brought to J26, because TinyUSB owns the native USB peripheral on GPIO19/20.

RFQ-EEG-001 Rev E E-28 asks for TP1-TP18 on the carrier plus exactly this 1x6 UART debug header at J26. The carrier provides both, so **there is no E-28 deviation**; the 2x5 1.27 mm JTAG/SWD header described in earlier drafts is withdrawn and no JTAG connector is fitted.

The DevKitC's two USB-C connectors are on the **non-isolated** side of the ADuM4160 barrier. They are used at the factory and on the bench only; the finished kit exposes only the isolator module's host USB-C (E-24, S-03), and T18 checks that the enclosure does not give a participant access to them.

2.4 `sdkconfig.defaults`, in full

This is the shipped file, verbatim. Every line is a requirement, not a preference.

Corrected 2026-09-02. The block below had stopped being verbatim. The five lines this section used to list under “not yet in the file and must be added before the first build” **are in the file now** – they were added on 2 September when the first real build refused to

proceed without them – and one of the five was named wrongly: there is no CONFIG_TINYUSB_VENDOR_ENABLED in esp_tinyusb 1.4.x, and ESP-IDF warns rather than fails on a key it does not recognise, so the setting looked present while CFG_TUD_VENDOR stayed 0 and the WebUSB interface did not exist. The key is CONFIG_TINYUSB_VENDOR_COUNT. The file is re-quoted here as it stands, with its own commentary, because a section headed “in full” that is not in full is worse than no quotation at all.

```
# FW-EEG-001 -- build configuration for EEG-CAR-01 Rev B.
# Every line here is a requirement, not a preference. Changing one
changes the instrument.
```

```
CONFIG_IDF_TARGET="esp32s3"
```

```
# ---- flash and PSRAM
```

```
-----
# The -N16R8 module has 16 MB quad flash and 8 MB OCTAL PSRAM. The
octal PSRAM uses
# GPIO35, 36 and 37, which is why those three pins are not connected on
the carrier
# (ECO-EEG-009).
CONFIG_ESPTOOLPY_FLASHSIZE_16MB=y
CONFIG_ESPTOOLPY_FLASHMODE_QIO=y
CONFIG_ESPTOOLPY_FLASHFREQ_80M=y
CONFIG_SPIRAM=y
CONFIG_SPIRAM_MODE_OCT=y
CONFIG_SPIRAM_TYPE_AUTO=y
CONFIG_SPIRAM_SPEED_80M=y
CONFIG_SPIRAM_USE_MALLOC=y
CONFIG_SPIRAM_MALLOC_ALWAYSINTERNAL=4096
```

```
# ---- partitions
```

```
-----
CONFIG_PARTITION_TABLE_CUSTOM=y
CONFIG_PARTITION_TABLE_CUSTOM_FILENAME="partitions.csv"
CONFIG_BOOTLOADER_APP_ROLLBACK_ENABLE=y
CONFIG_BOOTLOADER_APP_ANTI_ROLLBACK=y
```

```
# ---- USB
```

```
-----
----
# GPIO19 and GPIO20 are the USB PHY. Using TinyUSB in device mode
takes the PHY away
# from the USB-Serial-JTAG console, so UART0 on GPIO43/44 is the only
console -- which
# is why J26 exists.
CONFIG_TINYUSB_CDC_ENABLED=y
CONFIG_TINYUSB_CDC_COUNT=1
CONFIG_TINYUSB_DESC_USE_ESPRESSIF_VID=n
CONFIG_ESP_CONSOLE_UART_DEFAULT=y
CONFIG_ESP_CONSOLE_UART_NUM=0
```

```
# ---- radio: never initialised (RFQ E-18)
```

```
-----
CONFIG_ESP_WIFI_ENABLED=n
CONFIG_BT_ENABLED=n
```

```
# ---- security (RFQ F-19)
```

```
-----  
# Enabled for PHASE 2 ONWARD production builds only. The Phase 1  
# prototypes are built  
# with the sdkconfig.phase1 overlay and burn no eFuses. The signing  
# key never leaves the  
# programme; the manufacturer flashes a pre-signed image. See FW-  
# EEG-001 section 7.5.  
CONFIG_SECURE_BOOT=y  
CONFIG_SECURE_BOOT_V2_ENABLED=y  
CONFIG_SECURE_SIGNED_APPS_RSA_SCHEME=y  
CONFIG_SECURE_BOOT_BUILD_SIGNED_BINARIES=n  
CONFIG_SECURE_FLASH_ENC_ENABLED=y  
CONFIG_SECURE_FLASH_ENCRYPTION_MODE_RELEASE=y
```

```
# ---- timing
```

```
-----  
# The DRDY interrupt is the only place the sample counter moves (RFQ  
# E-19), so it must  
# be in IRAM and must not be delayed by a flash cache miss.  
CONFIG_FREERTOS_HZ=1000  
CONFIG_ESP_INT_WDT_TIMEOUT_MS=300  
CONFIG_ESP_TASK_WDT_TIMEOUT_S=10  
CONFIG_SPI_MASTER_ISR_IN_IRAM=y  
CONFIG_GPIO_CTRL_FUNC_IN_IRAM=y
```

```
# ---- storage
```

```
-----  
-  
CONFIG_FATFS_LFN_HEAP=y  
CONFIG_FATFS_MAX_LFN=255
```

```
# -----  
added to close FW-D11  
# FW-EEG-001 section 2.4 headed these "not yet in the file and must be  
# added before the  
# first build", and they were still absent. Without the first three  
# the vendor interface  
# and the custom descriptor set do not exist, so WebUSB does not  
# enumerate and the  
# _Static_assert on the MS OS 2.0 descriptor length never gets a chance  
# to fire.  
# CONFIG_TINYUSB_VENDOR_COUNT, not CONFIG_TINYUSB_VENDOR_ENABLED.  
#  
# esp_tinyusb 1.4.x has no option called TINYUSB_VENDOR_ENABLED, and  
# ESP-IDF does not fail  
# on a key it does not recognise -- it warns and carries on. So the  
# setting looked present,  
# CFG_TUD_VENDOR stayed 0, tusb.h declared none of the tud_vendor_*  
# functions, and the  
# WebUSB interface this device is built around would simply not have  
# existed. It surfaced  
# as "implicit declaration of function 'tud_vendor_mounted'" the first  
# time the firmware was  
# put in front of a real compiler.
```

```

CONFIG_TINYUSB_VENDOR_COUNT=1
CONFIG_TINYUSB_DESC_CUSTOM=y
CONFIG_TINYUSB_TASK_PRIORITY=18
CONFIG_BOOTLOADER_APP_SECURE_VERSION=1
CONFIG_BOOTLOADER_APP_ANTI_ROLLBACK=y
CONFIG_ESP_TASK_WDT_EN=y

```

CONFIG_SECURE_BOOT_BUILD_SIGNED_BINARIES=n is deliberate. The build container never holds the signing key; signing is a separate `espsecure.py sign_data` step performed on the custodian's offline machine (section 7.5).

CONFIG_SECURE_SIGNED_APPS_RSA_SCHEME selects RSA-3072 for secure boot v2. That is independent of the ATECC608B's P-256 device key: one authenticates firmware to the chip, the other authenticates recordings to the programme.

The five lines this section used to call missing are now present, and this is what became of each. *Corrected 2026-09-02: the table below used to be headed "lines that are not yet in the file and must be added before the first build".*

Line	Status	Note
CONFIG_TINYUSB_VENDOR_COUNT=1	added	FW-D03. This row used to name CONFIG_TINYUSB_VENDOR_ENABLED, which is not an option that exists; see above
CONFIG_TINYUSB_DESC_CUSTOM=y	added	FW-D02. It supplies the descriptors as data; the three callbacks <code>main.c</code> also defined had to be deleted , not kept
CONFIG_TINYUSB_TASK_PRIORITY=18	added	sits below <code>sample_task</code> at 23, as required
CONFIG_BOOTLOADER_APP_SECURE_VERSION=1	added	CONFIG_BOOTLOADER_APP_ANTI_ROLLBACK=y is meaningless without it
CONFIG_ESP_TASK_WDT_EN=y	added, and the subscription is not written	The option is on. Nothing subscribes <code>sample_task</code> or the SD path to the watchdog, so a stalled SD write still does not reset the unit. That half is open and is carried in section 10

Phase 1 overlay. The five security lines above are off for the two Phase 1 prototypes, which run **unsigned images and burn no eFuses at all**, so the firmware volunteer can re-flash them at the bench. Build Phase 1 images with `idf.py`

`-DSDKCONFIG_DEFAULTS="sdkconfig.defaults;sdkconfig.phase1"` build, where `sdkconfig.phase1` sets `CONFIG_SECURE_BOOT=n`, `CONFIG_SECURE_FLASH_ENC_ENABLED=n` and `CONFIG_BOOTLOADER_APP_ANTI_ROLLBACK=n`. **That file is now in the package**, as `firmware/sdkconfig.phase1`, and is listed in full below. Secure boot and flash encryption are enabled from Phase 2 onward; see section 7.5.

The overlay is the whole of the phase switch. It turns security options off and adds none, so a Phase 1 image and a Phase 2 image are the same firmware built two ways, and a Phase 1 prototype can be re-flashed at the bench for ever. `sdkconfig.defaults` is never edited to move between phases – editing it is how a prototype gets its eFuses burnt by accident.

`sdkconfig.phase1`, in full:

Corrected 2026-09-02. The quotation below had also stopped being verbatim, and the divergence was load-bearing rather than cosmetic. The file had acquired a second block of options appended after the ones quoted here, and one of them was `CONFIG_BOOTLOADER_APP_ANTI_ROLLBACK=y` – twenty lines below the deliberate `=n`. A later duplicate of the same key silently wins, so the Phase 1 intent was being undone by a line that read as an addition. It also made the build fail outright: ESP-IDF refuses anti-rollback on a partition table carrying a factory partition, and `partitions.csv` has one. That is how it was found – the first real ESP-IDF build stopped there. The file is re-quoted as it stands.

```
# FW-EEG-001 section 2.4 -- Phase 1 overlay for the two prototypes.
#
# Use: idf.py
# -DSDKCONFIG_DEFAULTS="sdkconfig.defaults;sdkconfig.phase1" build
#
# Phase 1 prototypes run UNSIGNED images and BURN NO eFUSES (RUL-
# EEG-021 section B), so the
# firmware volunteer can re-flash them over UART indefinitely. Do not
# put these lines in
# sdkconfig.defaults and do not build a Phase 2 unit with this overlay:
# a secure-boot-v2
# bootloader burns its eFuses on first boot, and with
# CONFIG_SECURE_FLASH_ENCRYPTION_MODE_RELEASE=y that unit can never be
# re-flashed over UART
# again. Phase 2 onward builds from sdkconfig.defaults alone; see
# section 7.5.
```

```
CONFIG_SECURE_BOOT=n
CONFIG_SECURE_BOOT_V2_ENABLED=n
CONFIG_SECURE_SIGNED_APPS_RSA_SCHEME=n
CONFIG_SECURE_BOOT_BUILD_SIGNED_BINARIES=n
CONFIG_SECURE_FLASH_ENC_ENABLED=n
CONFIG_SECURE_FLASH_ENCRYPTION_MODE_RELEASE=n
```

```
# Anti-rollback is an eFuse mechanism. With no eFuses burnt there is
# no anti-rollback and
# a downgrade is possible on a Phase 1 prototype; that is accepted and
# stated in section 7.5.
CONFIG_BOOTLOADER_APP_ANTI_ROLLBACK=n
```

```
# -----
# added to close FW-D11
# FW-EEG-001 section 2.4 headed these "not yet in the file and must be
# added before the
# first build", and they were still absent. Without the first three
# the vendor interface
# and the custom descriptor set do not exist, so WebUSB does not
# enumerate and the
```

```

# _Static_assert on the MS OS 2.0 descriptor length never gets a chance
to fire.
# CONFIG_TINYUSB_VENDOR_COUNT, not CONFIG_TINYUSB_VENDOR_ENABLED.
#
# esp_tinyusb 1.4.x has no option called TINYUSB_VENDOR_ENABLED, and
ESP-IDF does not fail
# on a key it does not recognise -- it warns and carries on. So the
setting looked present,
# CFG_TUD_VENDOR stayed 0, tusb.h declared none of the tud_vendor_*
functions, and the
# WebUSB interface this device is built around would simply not have
existed. It surfaced
# as "implicit declaration of function 'tud_vendor_mounted'" the first
time the firmware was
# put in front of a real compiler.
CONFIG_TINYUSB_VENDOR_COUNT=1
CONFIG_TINYUSB_DESC_CUSTOM=y
CONFIG_TINYUSB_TASK_PRIORITY=18
CONFIG_BOOTLOADER_APP_SECURE_VERSION=1
# NOT CONFIG_BOOTLOADER_APP_ANTI_ROLLBACK=y here. It was appended in
this block and it
# contradicted the deliberate `=n` twenty lines above, which explains
why a Phase 1
# prototype has no anti-rollback: it is an eFuse mechanism and no
eFuses are burnt. A
# later duplicate of the same key silently wins, so the Phase 1 intent
was being undone by
# a line that looked like an addition. It also makes the build fail
outright -- ESP-IDF
# refuses anti-rollback on a table that has a `factory` partition, and
partitions.csv has
# one -- which is how it was found: the first real ESP-IDF build
stopped here.
CONFIG_ESP_TASK_WDT_EN=y

# ----- IRAM
headroom
# The first real build came back with static IRAM at 100.0 % -- 16,383
of 16,384 bytes,
# ONE byte free. That is a cliff, not a pass: the next function anyone
marks IRAM_ATTR
# fails the link with an error naming a section rather than a cause,
and this design has
# more interrupt work coming (the E-13 tone scheduler, the E-12 onset
detector).
#
# Two ISRs are kept in IRAM by default and neither is used by this
firmware. There is no
# SPI slave anywhere in the design -- the ESP32-S3 is the master and
both ADS1299 modules
# are slaves -- and nothing uses gptimer; `grep spi_slave` and `grep
gptimer` over main/
# return nothing. Reclaiming them costs this design nothing.
#
# SPI MASTER stays in IRAM deliberately: sample_task() reads the
converters on every DRDY

```

```
# and that path must survive a flash-cache stall.
CONFIG_SPI_SLAVE_ISR_IN_IRAM=n
CONFIG_GPTIMER_ISR_HANDLER_IN_IRAM=n
#
# MEASURED AFTERWARDS, AND IT DID NOT HELP. The figure is still 16,383
of 16,384 with one
# byte free, byte for byte, so whatever fills that pool it is not these
two ISRs. The
# change is kept because carrying interrupt handlers for a bus and a
timer this firmware
# does not use is wrong either way, but it is NOT the fix and must not
be mistaken for one.
# Either the pool is genuinely full and a bring-up engineer has to
choose what leaves IRAM,
# or esp_idf_size is reporting against a fixed 16 KB window that is not
the real limit on
# an ESP32-S3 with octal SPIRAM and XIP. Deciding which needs the
linker map read against
# hardware, so it is carried as an open item rather than guessed at
from here.
```

The three security lines named in the paragraph above are the ones that matter; the other four are their dependants, set explicitly so that `idf.py` reports the same configuration on every bench rather than resolving them silently.

Two blocks follow them and neither is a security line. The first repeats the FW-D02 / FW-D03 / FW-D11 options of the previous subsection; an overlay build reads `sdkconfig.defaults` first and then this file, so those four lines are **redundant rather than load-bearing** – they are kept because a reader of this overlay alone should be able to see the whole of what a Phase 1 image is built with. The second is the pair of IRAM lines, `CONFIG_SPI_SLAVE_ISR_IN_IRAM=n` and `CONFIG_GPTIMER_ISR_HANDLER_IN_IRAM=n`: an attempt at the open item of section 9 that **did not work**, is kept for a different reason than the one it was made for, and says so in the file.

sdkconfig.qemu – an emulator overlay, and not a configuration for any unit. Added 2 September 2026 so the firmware could be run without hardware (section 8.2). It differs from the shipped build in exactly one option, and that option is why the QEMU boot gets as far as it does:

```
# FW-EEG-001 -- overlay for running the firmware under QEMU.
#
# NOT a build configuration for any real unit. It exists so the
firmware can be exercised
# without hardware, and it differs from the shipped build in exactly
one way that matters.
#
# QEMU's esp32s3 machine does not emulate octal PSRAM. The shipped
configuration aborts
# when PSRAM is absent -- "Failed to init external RAM!" -- and the
unit then reboots in a
# loop. That is defensible on real hardware, because the 6 MiB ring
buffer of F-06 lives
# in PSRAM and a unit without it cannot do its job; it is also, on a
field device, a
# failure mode that says nothing to anyone. A unit that boot-loops in a
```

```

participant's home
# cannot be told apart from a dead battery or a bad cable, and the
browser tool cannot ask
# it what is wrong because it never gets far enough to enumerate. See
KNOWN_ISSUES.
CONFIG_SPIRAM_IGNORE_NOTFOUND=y

```

Never build a unit with this overlay. CONFIG_SPIRAM_IGNORE_NOTFOUND=y turns a missing 6 MiB ring buffer – which is F-06, and the whole of the retransmit story – from a refusal to boot into a silent degradation, and a field unit that records without a ring is a unit that loses data the host asked it to keep.

2.5 partitions.csv, in full

FW-EEG-001 partition table -- A/B OTA with rollback (RFQ F-20), 16 MB flash.

```

# Name,      Type, SubType,  Offset,    Size,      Flags
nvs,         data, nvs,        0x9000,    0x6000,
otadata,     data, ota,        0xf000,    0x2000,
phy_init,    data, phy,        0x11000,   0x1000,
factory,     app,  factory,    0x20000,   0x300000,
ota_0,       app,  ota_0,       0x320000,  0x300000,
ota_1,       app,  ota_1,       0x620000,  0x300000,
calib,       data, nvs,        0x920000,  0x8000,
prov,        data, nvs,        0x928000,  0x8000,
storage,     data, fat,      0x930000,  0x6D0000,

```

Partition	Size	Contents
nvs	24 kB	general non-volatile settings
otadata	8 kB	two 4 kB sectors selecting the boot slot
phy_init	4 kB	present but unused; the radio is never initialised (E-18, S-06)
factory	3 MB	the recovery image flashed at end of line and never overwritten in the field
ota_0, ota_1	3 MB each	the A/B pair of F-20
calib	32 kB, NVS namespace eegcal	the constants written at T6 from T7, T10, T12 and T17
prov	32 kB, NVS namespace eegcfg	USB PID, hardware revision, unit serial, ATECC serial, provisioning timestamp

Corrected 2026-09-02. This table was right and the firmware did not implement it. drivers.c opened namespace "tioV" in the **default** nvs partition for all four keys, so the two partitions above were allocated at 0x920000 and 0x928000 and never written – 64 kB reserved and unused – while provisioning and calibration went into the default partition instead. That partition is 0x6000 = **24 kB** and is shared with system state, so the 32 kB calibration set this table specifies could not have fitted in it: a full-size calibration write would have failed with an out-of-space error naming a partition no

document said was involved. `drv_nvs_set_str()`, `drv_nvs_set_blob()` and `unit_serial_into()` now open `calib/eegcal` for the calibration blob and `prov/eegcfg` for identity, initialising each partition on first use and erasing only that partition if it has never been written.

Reading the calibration back is now possible. `CMD_READ_CALIBRATION (0x4A)` returns a slice of the stored blob – payload `offset_lo offset_hi length` – so a host can compare byte for byte with what it sent. TST-EEG-004 T6’s acceptance limit is that the unit reads back what was written, and until this opcode existed nothing could read it back at all. It is chunked because a full calibration set does not fit in one frame, and it is exempt from the provisioning-mode gate because T6 runs after provisioning, on a unit that has left that mode and may have had its configuration zone locked. | storage | 6.81 MB, FAT | block audio (F-11) and the 1 kHz reference tone (E-16, F-21) |

Arithmetic check: $0x930000 + 0x6D0000 = 0x1000000 = 16 \text{ MB}$ exactly, so the table fills the device with no overlap.

Measured, 2 September 2026. *This paragraph used to read “a reference application build is expected at roughly 1.1 MB ...* **Calculated, not measured** – *nothing has been built.”* It has been built. `idf.py size` reports a **405,245-byte** image; `eeg_field_kit.bin` in `firmware/release/` is **405,360 bytes** on disk, the difference being the padding and the appended SHA-256 that `esptool` writes into the flashable file. Against a 3 MB slot that is **13 %** used, so the headroom for the FAT, I2S and SDMMC code the drivers still need is large. The QEMU boot log shows the bootloader loading exactly this table and taking the app from factory at `0x20000`, so the offsets in it are not merely arithmetic any more.

The one memory figure that is **not** comfortable is static IRAM: 16,383 of 16,384 bytes used, one byte free. That is section 10 item 17 and it is open.

At 48 kHz, 16-bit mono, the 6.81 MB FAT partition holds about **71 seconds** of audio (calculated). If a block’s stimulus audio is longer than that it must be served from the microSD card instead. That decision is open and is listed in section 10.

3. The pin map

`main/board_pins.h` is **generated** from `tools/design.py` so that the firmware and the board cannot disagree. Do not edit it by hand: edit `design.py` and regenerate.

The GPIO map itself lives in ICD-EEG-006 section 5, as the J6 and J7 way tables, and is not restated here. Read the GPIO number and the header way for any net there. What this document owns is the set of macro names the generated header defines, and the four constraints below that a firmware author has to know.

Macro	Net	Macro	Net
<code>PIN_SR_LATCH</code>	<code>LED_SR_LATCH</code>	<code>PIN_I2S_BCLK</code>	<code>I2S_BCLK</code>
<code>PIN_I2C_SDA</code>	<code>SDA</code>	<code>PIN_USB_DN</code>	<code>USB_DN</code>
<code>PIN_I2C_SCL</code>	<code>SCL</code>	<code>PIN_USB_DP</code>	<code>USB_DP</code>
<code>PIN_ENV_CMP</code>	<code>ENV_CMP</code>	<code>PIN_MIC_MUTE</code>	<code>MIC_MUTE</code>

Macro	Net	Macro	Net
PIN_BTN_A	BTN_A	PIN_SD_CMD	SD_CMD
PIN_BTN_B	BTN_B	PIN_SD_CLK	SD_CLK
PIN_BTN_STOP	BTN_STOP	PIN_SD_D0	SD_D0
PIN_I2S_DIN	I2S_DIN	PIN_SR_DATA	LED_SR_DATA
PIN_I2S_LRCK	I2S_LRCK	PIN_SR_CLK	LED_SR_CLK
PIN_I2S_DOUT	I2S_DOUT	PIN_UART0_TX	UART_TX
PIN_ADS_CS	CS	PIN_UART0_RX	UART_RX
PIN_ADS_MOSI	MOSI	PIN_VBUS_DET	VBUS_DET
PIN_ADS_SCLK	SCLK	PIN_CHG_CE	CHG_CE
PIN_ADS_MISO	MISO	PIN_LED_V	LED_PWM
PIN_ADS_DRDY	DRDY	PIN_RESERVED_PSR AM_D5/D6/D7	not connected
PIN_ADS_START	START	PIN_RESERVED_VDD _SPI	not connected
PIN_ADS_RESET	RESET		

ECO-EEG-009. The Rev A map put LED_SR_DATA, LED_SR_CLK and LED_SR_LATCH on GPIO35, 36 and 37. On the ESP32-S3-DevKitC-1-N16R8 those three pins carry the **octal PSRAM**; they are not available as user IO and the firmware pin map was unbuildable. Rev B moves the shift register to GPIO41, 42 and 0 and drops the microSD interface to one-bit SDMMC to free them. The sustained microSD write requirement of E-20 is about 70 kB/s – the 50.7 kB/s of frame payload calculated in section 5.8 plus STATUS and SIGNATURE frames and filesystem overhead – against about 2 MB/s available in one-bit mode, so the headroom is ample (calculated). GPIO45 is the VDD_SPI strapping pin and is left open; a pull-up on it would set VDD_SPI to 1.8 V and the module would not boot.

main.c carried the Rev A defines until 2 September 2026, and a build of that source would have produced a unit whose contact lights did not work and whose PSRAM was corrupted by lights_write(). The block is deleted and main.c:48 includes the generated board_pins.h, so FW-D12 is closed; the aliases that follow the include exist only so the rest of the file keeps its original names. The generated header is the only pin map, and it comes from design.py: regenerate it, never edit it, and the firmware and the board cannot disagree.

board_pins.h also carries N_CH 16, N_DEV 2, ADS_GAIN_EEG 24, ADS_GAIN_EMG 12, ADS_GAIN_ENV 1, N_LIGHTS 8 and LIGHT_PHASE_HZ 240, and documents the dark-at-boot guarantee: LED_V is GPIO48, an input at reset, so no current can flow through any light whatever the shift register contains (E-27). LIGHT_PHASE_HZ is the **nominal** figure; the driver quantises it to the FreeRTOS tick and runs at about 250 Hz (section 3.2).

Verified by a build with -Werror plus T11 (contact-light colour in all three states) and T3 (board boots and draws expected current). *Status 2026-09-02: the build half is done –*

ESP-IDF v5.2.5, clean at -Wall -Werror=all, with board_pins.h compiled into the released image. T11 and T3 both need a unit and neither has been run. **Signed off by** the firmware volunteer against the generated header, with the generation re-run recorded in the release manifest – that sign-off has not happened.

3.1 What the layout changed, and what it means for the firmware

Two things changed in the carrier during layout, and the firmware author needs to know both even though neither moves a pin.

The carrier is 150.0 x 130.0 mm, not the 130 x 124 mm of package v1. Thirty connectors, 211 parts and 156 nets would not close at the smaller size.

The carrier is a four-layer board: L1 signal, L2 reference plane, L3 reference plane, L4 signal. Package v1 asserted that two layers would be enough and cheap to route. Actually doing the layout showed that it is not: on two layers the bottom side has to be both the reference plane and the second routing surface, and it cannot be both. Four layers give two full routing surfaces and a continuous reference under every analogue trace, which is what DSN-EEG-002 section 13's "layout rules that are requirements, not preferences" ask for and a swiss-cheesed two-layer pour cannot deliver. Both inner layers carry AGND_REF left of $x = 62$ mm and DGND right of it, tied at the R90 star point only, which is the zoning and star-point rule of DSN-EEG-003 section 3.3; the isolation keep-out of that same section – $x \geq 141$ mm, $y = 2$ to 22 mm, no copper on any layer – is where the harness barrier crosses the board. Vias are through vias only, 0.60 mm pad on a 0.30 mm finished hole; there are no blind, buried, back-drilled, filled or plugged vias. The stack-up is mask / 35 μ m L1 / prepreg 0.200 / 17 μ m L2 / core 1.065 / 17 μ m L3 / prepreg 0.200 / 35 μ m L4 / mask, 1.60 mm +/- 10 % finished. The enclosure grew with the board: POD-P1 base 163.0 x 143.0 x 58.0 mm external and 158.0 x 138.0 x 55.5 mm internal, MP-01 module plate 146.0 x 126.0 x 3.0 mm.

The routing closes, and it is released for review rather than for fabrication. The firmware author will be handed a board file that may still change, and should know why. kicad/EEG-CAR-01_RevB_DRC_report.txt is the authority. It records 3 745 track segments and 552 through vias on the four layers, each reference plane one continuous island per net, a smallest measured clearance of 0.260 mm on L1, 0.285 mm on the two planes and 0.275 mm on L4 against a 0.20 mm rule, a narrowest conductor of 0.20 mm, a smallest plated hole of 0.30 mm, all 145 nets one connected copper island with none left without copper, no digital net anywhere in the analogue zone, exactly one AGND_REF-to-DGND bridge and exactly one HARN_SHIELD-to-DGND bridge, and the isolation strip clear of copper on all four layers – that last one because the report says so, not because it was assumed. It records **zero violations** – no clearance, width, annular-ring, hole-size, edge, non-plated-hole, isolation or via keep-out violation, and no unclosed connection. It also records that **169 connections were routed at relaxed geometry**: 36 took a conductor narrower than the 0.25 mm preferred width and 133 kept full width but took a reduced gap, all of them at or above the 0.20 mm minimum. **The fabrication data is RELEASED FOR REVIEW under RFQ-EEG-002A and is not released for fabrication**, because no human layout engineer has read routing produced by the programme's own tools, and a board that closes at minimum geometry is not the board that closes at preferred geometry. Nothing here changes a pin assignment, so board_pins.h and every table in this document stand as written.

The firmware consequence is narrow and it is procedural: board_pins.h is generated from design.py, so every geometry change obliges a regeneration and a diff of the header against the previous release before the next build, and the release manifest

records that the regeneration was run. The board field in the provisioning record (section 7.4) stays EEG-CAR-01 Rev B, which is the electrical revision; the geometry change is carried as ECOs against it and does not bump bcdDevice.

3.2 The contact-light phase scheme, now written

Each of the eight sites carries a two-lead bicolour LED between its shift-register output Qn and the LED_V common on GPIO48, with a 1 kOhm series resistor R70-R77. With $V_f = 2.0\text{ V}$ the per-site current is $(3.3 - 2.0)/1000 = \mathbf{1.3\text{ mA}}$, and all eight together draw **10.4 mA** from GPIO48. Phase A (LED_V high, Qn low) shows green, phase B (LED_V low, Qn high) shows red, and alternating the two phases faster than the eye can follow shows amber. The requirement is that the alternation runs above 100 Hz; the fitted value is LIGHT_PHASE_HZ 240.

Corrected 2026-09-02. This subsection used to be headed “and the fact that it is not written”, and said “**None of this is implemented** ... TST-EEG-004 T11 ... cannot pass until the phase driver is written”. **It is written now**, in lights_phase() and lights_task() at main.c:325-360, and FW-D16 is closed. **E-27 is met in the design and in the source.** It has never been run: no unit exists, and QEMU has no shift register.

Three things about the implementation belong here because they are not obvious from the scheme above.

The colour comes from the converter’s lead-off comparator, read at two thresholds. A site that trips neither is **green**, a site that trips only the sensitive threshold is **amber** – marginal, re-gel it – and a site that trips both is **red**. That is the three-state indication E-27 asks for, taken from the measurement E-27 names, rather than a colour chosen by firmware state. The insensitive set is a subset of the sensitive one by construction, so the three states are exhaustive and cannot overlap.

Corrected again 2026-09-02 (FW-D17). This paragraph read “**The colour comes from both halves of the converter’s lead-off measurement.** The driver reads LOFF_STATP and LOFF_STATN, not one of them.” It did, and that was the defect. ads_init() enabled LOFF_SENSP only, so LOFF_STATN was zero on every channel for ever, the red term p & n was always zero, and **red was unreachable** – every site that had lost contact showed amber. Enabling LOFF_SENSN would not have fixed it: this board’s montage is **single-ended**, J2 carrying IN1 to IN8 with one shared SRB1 and BIASOUT, and with SRB1 closed all eight N bits report that one shared reference rather than eight sites. Sweeping COMP_TH is what the hardware actually supports. The two threshold values are the ADS1299’s documented endpoints, not measured trip points, so **the impedance at which a site turns amber, and the impedance at which it turns red, are not yet established** – TST-EEG-004 T11 sets them at first bring-up.

The alternation is about 250 Hz, not 240. LIGHT_PHASE_HZ is 240, so a half-phase is 2.083 ms; the task delays in FreeRTOS ticks and CONFIG_FREERTOS_HZ is 1000, so the half-phase quantises to 2 ms and the alternation runs at **250 Hz**. This is stated rather than hidden. The requirement is “above 100 Hz” and 250 Hz meets it with room; both half-phases quantise identically so the duty stays 50/50 and the colour does not shift; and T11 reads the red/green ratio with a colorimeter, which does not care about 4 % of frequency. A build that needs exactly 240 Hz needs a hardware timer rather than a task delay, and that is different work.

Dark is dark by construction. While a block is recording, or when the host has switched the lights off, lights_task() clears the shift register **and returns LED_V to**

an input – the state it holds at reset. With the common floating no current can flow through any site whatever the register contains. Driving LED_V low would also be dark, but only for as long as the level is right.

CMD_LIGHTS still does not do what section 6.3's argument column describes, and this paragraph is corrected too: it used to say the source “answers 0x0B, not implemented, to all three” of modes 2, 3 and 4. It does not. `handle_command()` takes the mode byte as a plain enable – `g_lights_enabled = c[1]` – and answers 0x00 OK to any value, while the colour is decided by the lead-off measurement rather than by the mode. So a host asking for “force red” gets a success and the automatic colour, which is a worse answer than a refusal. Forcing a colour for T11 is not implemented; the honest reply would be 0x0B and the source does not send it. That is carried in section 10.

4. The USB device model

4.1 Composite configuration

Item	Value
bcdUSB	0x0210 – required for a BOS descriptor to be read
Device class	TUSB_CLASS_MISC / MISC_SUBCLASS_COMMON / MISC_PROTOCOL_IAD
Speed	full speed, 64-byte packets
Interface 0, 1	CDC-ACM: notification EP 0x81 (8 B), data OUT 0x02, data IN 0x82 (F-01, WebSerial path)
Interface 2	vendor-specific bulk: OUT 0x03, IN 0x83, 64 B (F-01, WebUSB path)
bMaxPower	100 mA declared; the instrument is battery powered and draws nothing from the host
Strings	1 “TI One Voice”, 2 “EEG field kit”, 3 iSerial, 4 “EEG CDC”, 5 “EEG WebUSB”

Both interfaces carry the **identical** frame stream. Whichever the host opens first wins; if both are open, `frame_emit()` writes to both.

4.2 BOS, WebUSB, Microsoft OS 2.0

The BOS descriptor declares two platform capabilities.

Capability	Vendor request	Serves
WebUSB platform	<code>bRequest = 1</code>	landing URL https://one.witysk.org/ee g, scheme byte 1 = https
Microsoft OS 2.0 platform	<code>bRequest = 2, wIndex = 7</code>	a 178-byte (0xB2) descriptor set

The MS OS 2.0 set assigns the compatible ID WINUSB to the vendor interface and sets the registry property DeviceInterfaceGUIDs, so Windows binds WinUSB with no user action (F-02). The length is asserted at compile time:
_Static_assert(sizeof(ms_os_20_desc) == MS_OS_20_DESC_LEN, ...) at main.c:705. *Corrected 2026-09-02: this read “That assert has never been evaluated (FW-D11); it is the test, and it fires on the first build.”* The first build happened, the assert was evaluated against the resolved esp_tinyusb 1.4.5 macros, and it **held**. FW-D11 is closed. What that settles is the descriptor’s length, not whether Windows binds WinUSB to it; that is T5.

The GUID in the shipped source, {8FE6D4D7-49DD-41E7-9486-49AFC6BFE475}, is the TinyUSB example GUID. It must be replaced with a GUID minted once for this programme and recorded in the release manifest, otherwise Windows may bind any other TinyUSB sample device on the same PC to the same interface class.

4.3 VID and PID

Where	Value	Status
main.c:673-674	0x1209 / 0x0000	0x0000 is not a valid product ID. <i>Line reference corrected 2026-09-02</i>
provision.py DEFAULT_PID	0x0001	pid.codes reserves 1209:0001 for testing only
Allocated	pending	pid.codes application under VID 0x1209

F-03 and F-18 say the manufacturer programs the assigned VID and PID at end of line, but main.c compiles them into a static const tusb_desc_device_t, so nothing can be programmed. **Decision, Rev B, unchanged:** idProduct, bcdDevice and iSerialNumber are read from NVS namespace eegcfg at boot, **before** tinyusb_driver_install(), with compiled-in fallbacks. A unit that boots on the fallback must enumerate as iProduct = “EEG field kit UNPROVISIONED” so that T5 fails loudly rather than shipping. bcdDevice = 0x0100 corresponds to hardware revision B.

The two Phase 1 prototypes may carry 1209:0001 with a written note that they are never given to a participant. Browser authorisation is keyed to VID, PID and the serial string, so the permission grant has to be re-made once the real PID lands. That is acceptable for two prototypes and unacceptable for a fleet.

5. Frame format as an implementation contract

RFQ-EEG-001 section 5.2 gives the field list. This section gives the bytes. All multi-byte integers are **little-endian** unless stated. Every frame crosses every boundary unchanged: microSD, USB and the browser-to-server websocket. One parser, one test suite.

5.1 Envelope

Offset	Size	Field	Notes
0	1	version	1 for this document. A host that sees another value must reject the frame, not parse it
1	1	type	1 DATA, 2 STATUS, 3 EVENT, 4 GAP, 5 SIGNATURE, 6 CMD_ACK, 16 CMD (host to device)
2	2	seq	u16, increments per frame, wraps at 0xFFFF
4	4	first_sample	u32, the DRDY counter of the first sample. The only timeline in the system (E-19)
8	1	rate_code	0 = 250 Hz, 1 = 500 Hz, 2 = 1000 Hz
9	1	n_samples	20 at 1000 Hz, 10 at 500, 5 at 250; 0 for non-DATA frames
10	n	payload	per type, below
10+n	4	crc32	IEEE 802.3 over bytes 0 to 10+n-1

The whole of the above is then COBS-encoded and a single 0x00 delimiter is appended. A decoder joining mid-stream resynchronises at the next 0x00, which is what the delimiter is for. Header struct in Python: `struct.Struct("<BBHIBB")`, size 10.

5.2 DATA payload

`n_samples` records of exactly **50 bytes**:

Offset in record	Size	Field
0	48	channels 1 to 16, each 24-bit two's complement, big-endian , in ADS1299 output order
48	2	auxiliary field, u16 little-endian

Channel map, normative:

Channel	Signal	Gain	Source
1-8	Fz, Cz, Pz, C3, C4, T7, T8, F7	24	ADS module #1, J14

Channel	Signal	Gain	Source
9-11	EMG cheek, submental, laryngeal	12	ADS module #2 ch1-3, J15-J17
12	ENV_STIM	1	ADS module #2 ch4, J4.4
13	ENV_VOICE	1	ADS module #2 ch5, J4.5
14	ENV_ROOM	1	ADS module #2 ch6, J4.6
15-16	spare / EOG	24	ADS module #2 ch7-8, J22

Channels 15 and 16 exist in the stream on every unit and are protected like every other electrode lead, but **the EOG panel sockets are not fitted in a standard build**; they are a Phase 2 option. The two channels record whatever J22 is left presenting.

Channel 12 is the authoritative stimulus onset record (F-11). Note that this numbering is the *stream* numbering and is not the same as the R1-R16 protection-network numbering in DSN-EEG-003, which is numbered by electrode lead and includes REF_L, REF_R and BIAS. The two lists are different things and must not be conflated.

A reversed daisy chain inverts the whole map. It is detected at T7 because a channel 9 reading gain 24 rather than 12 can only mean the modules are the other way round – see T27.

5.3 Auxiliary field

Bit	Meaning	Source
0	BTN_A pressed	GPIO4, active low
1	BTN_B pressed	GPIO5
2	BTN_STOP pressed	GPIO6
3	stimulus comparator, latched in the DRDY ISR	GPIO3, E-12
4	any lead-off asserted	LOFF_STATP
5	recording block in progress	firmware state
6	charger input present	GPIO46 VBUS_DET
7	microSD write healthy	SD driver
8-11	reserved, zero	
12-15	protocol reserved, zero	

`main.c` sets bits 0 to 5 and samples bit 3 in the task rather than in the ISR (FW-D08).

Bit 3 reads GPIO3, the output of the U7 comparator. ECO-EEG-023, which is **open and not implemented**, would re-power U7 from DVDD3V3 and DGND and re-reference its inputs to a DVDD3V3/2 divider with the envelope AC-coupled into it, so that GPIO3 would swing a full 0 to 3.3 V. As released, U7 is on AVDD/AVSS and the clamp leaves 20 mV of logic margin. That is a change the safety and layout reviewer must check; the firmware side is unaffected beyond the improved margin.

5.4 STATUS payload – 24 bytes, once per second (F-09)

Offset	Size	Field
0	1	battery percent, 0-100
1	1	link type: 0 none, 1 CDC, 2 vendor
2	2	lead-off bits, channels 1-16
4	4	microSD free space, MB
8	4	uptime, seconds
12	2	die temperature, i16, tenths of a degree C
14	4	clock-offset estimate, i32, microseconds. Metadata only; never applied to the sample timeline (F-16)
18	2	ring depth actually allocated, seconds – T15 records this rather than assuming it
20	3	firmware version, major / minor / patch
23	1	flags: b0 recording, b1 charging inhibited, b2 card present, b3 provisioned

The die temperature at offset 12 is the ESP32-S3 internal sensor. It is **not** a battery temperature: there is no NTC net on the carrier and no thermistor way on J12 or J13, so the thermistor-monitored charging of RFQ S-04 is **not met and stays not met** as an open hardware item. The 45 degree C charge inhibit of RFQ E-23 is a property of the charger IC's own thermal regulation, not of anything the firmware reads.

5.5 GAP payload – 10 bytes (F-07)

Offset	Size	Field
0	4	first sample index lost
4	4	last sample index lost
8	2	frames lost

Silent loss is not permitted. `verify_stream.py` resumes its sample-index expectation at `last + 1` when it sees a GAP, which is the only way a legitimate loss is distinguished from a decoder fault. A GAP is also the marker the host uses to decide which range to recover from the microSD copy, which is now part of how F-06 is met – section 5.8.

5.6 SIGNATURE payload – 104 bytes, every 2048 samples (F-08)

Offset	Size	Field
0	64	ECDSA P-256 signature, raw r \ \ s, 32 bytes each, big-endian
64	4	block index, u32
68	4	first sample index of the block
72	32	block hash

The chain, exactly as `verify_stream.py` computes it: `block_hash` is SHA-256 over the concatenation of every DATA frame body in the block **excluding its CRC**; the signed digest is SHA-256(`previous_chain_value` || `block_hash`); the chain value becomes that digest. The signature is computed by the ATECC608B over the digest. Datasheet ATECC sign time is typically 50 to 70 ms; at 1000 Hz a block is 2.048 s, so the margin is large, but signing must not run on the sample task. Nothing has been measured on hardware.

5.7 EVENT payload

Offset	Size	Field
0	1	event code: 1 block start, 2 block end, 3 button, 4 host mark
1	1	label length, 0 to 32
2	n	label, UTF-8, maximum 32 bytes

5.8 Bandwidth and ring depth – calculated

One frame at 1000 Hz is 10 header + 20 x 50 payload + 4 CRC = **1014 bytes**, which is 1015 bytes after COBS encoding and 1016 bytes on the wire once the delimiter is added. Fifty frames per second is therefore **50.7 kB/s of frame payload** – the figure every document in this package now uses – and about 50.8 kB/s on the wire.

E-20's "approximately 70 kB/s" and F-12's "approximately 64 kB/s" are not in conflict with that number and neither requirement changes. They are **allowances**: they include the once-per-second STATUS frames, the SIGNATURE frame every 2.048 s, and FAT filesystem overhead on the microSD copy, none of which is frame payload. 70 kB/s remains the microSD sizing number and the E-20 acceptance figure.

Ring depth. F-06 as originally written asks for at least three minutes of ring at 1000 Hz and sizes that at **12 MB**. The 12 MB is not a mistake and this document no longer calls it one: it is three minutes at E-20's 70 kB/s allowance, 180 x 70 kB = 12.6 MB rounded down, which is the figure DSN-EEG-003 section 5, TST-EEG-004 T15, RFQ-EEG-001 F-06 and the ECO-EEG-025 title all quote. Measured instead against the frame payload derived above, three minutes at 50.7 kB/s is **9.13 MB** (calculated). The two arithmetics differ because one counts the allowance and the other counts the payload, and the difference does not matter here, because the ESP32-S3-DevKitC-1-N16R8 (E-18, not substitutable) has **8 MiB** of PSRAM in total, before any heap. Neither 12 MB nor 9.13 MB can be allocated: `main.c` asked for 12 MiB, that allocation returned NULL, and every subsequent `xRingbufferSend()` dereferenced it. That was FW-D13 and it is

fixed – RING_BYTES is 6 MiB at `main.c:101`, and `app_main()` now aborts with a diagnostic naming the PSRAM configuration rather than carrying on if the allocation ever fails again.

Ruling, carried as ECO-EEG-025. RING_BYTES = 6,291,456 bytes – a **6 MiB** ring, and the “6 MB ring” the rest of the package names.

Figure	Value	How it is counted
Allocation	6,291,456 bytes	RING_BYTES, 6 MiB
Depth at 1000 Hz, as the simulator reports it	126 s	<code>tools/simulate_production.py</code> , over the 50 bytes per sample the ring is sized to hold
Depth at 1000 Hz, over the complete framed stream	124 s	50.7 kB/s from the paragraph above, header, CRC and COBS included
Depth at 500 Hz, over the complete framed stream	244 s	one frame every 20 ms carrying ten samples
Requirement, F-06 as relaxed by ECO-EEG-025	90 s	plus unlimited backfill from the microSD copy

`simulate_production.py` grades the check “F-06 as relaxed by ECO-EEG-025: at least 90 s of ring” and passes it at 126 s. Whichever of the two arithmetics is used the fitted ring is more than a third deeper than the relaxed requirement, so the choice between them changes no decision; both are given so that no reader has to guess which one produced a number quoted elsewhere.

F-06 is relaxed to 90 seconds of ring at 1000 Hz plus unlimited backfill from the microSD copy. The reason is worth stating: the ring exists to survive a USB stall or a browser tab losing focus, and the card holds the same byte stream in full, so anything older than the ring is recovered from the card rather than from PSRAM. Ninety seconds of live retransmit plus a complete card copy is a better answer than three minutes of PSRAM the module does not have. The device declares the depth it actually allocated in the STATUS frame and T15 grades against the declared value, not against any figure in the table above. The alternative – a 16 MB-PSRAM module – contradicts E-18 and is not recommended.

5.9 The microSD file layout

This is the definition of what is on the card, and it is made once, here. SVC-EEG-013 section 2 R2, TST-EEG-004 T14 and the programme’s ingest tooling cite this section and do not restate it. The earlier SVC-EEG-013 layout – `/SESSIONS/<unit_serial>/<YYYYMMDD>T<HHMMSS>Z_<session_id>.eegs` with a 512-byte plain-text header – is withdrawn, for two reasons that are worth writing down. The unit has no real-time clock, so at the moment a file is created the device may not yet know the UTC time; it learns it from `CLOCK_XCHG`, which the host may send late or not at all, and a filename that cannot always be formed is not a filename rule. And an in-file header breaks the one contract section 5 exists to state: every frame crosses every boundary unchanged, so the card holds exactly the bytes that went to USB and one

parser reads both. The UTC start time the service engineer wants is kept, in the sidecar, where a missing clock exchange leaves a null instead of an unusable name.

Item	Definition
Filesystem	exFAT , formatted by the programme before the card is issued. The firmware never formats a card
Volume label	TI0V<nnnn>, the four serial digits, so a card found loose can be returned to its unit
Directory	/EEG/<unit_serial>/, one directory per unit, <unit_serial> exactly the TI0V-B-nnnn string of section 7.2
Session file	<session_id>.eeg, where <session_id> is the 16 bytes of START_SESSION rendered as 32 lowercase hexadecimal characters
Session file content	The byte-identical COBS frame stream that went to USB: DATA, STATUS, EVENT, GAP and SIGNATURE frames, in the order emitted, each with its 0x00 delimiter. No in-file header, no index, no padding, no trailer
Sidecar	<session_id>.json, UTF-8, written at STOP_SESSION
Provisioning record	/EEG/<unit_serial>/unit.json, the section 7.4 identity block, written once at provisioning and never rewritten by a session

The sidecar carries what a reader cannot recover cheaply from the stream, and nothing that contradicts it:

Field	Type	Notes
schema	u8	1 for this document
unit_serial	string	as on the label
session_id	string	32 hex characters, matching the filename
board	string	EEG-CAR-01 Rev B
firmware_version	string	and image_sha256, both as GET_VERSION reports them
utc_start	string or null	ISO 8601, derived from the last CLOCK_XCHG before the first frame. Null if the host never sent one , and a null is not an error

Field	Type	Notes
rate_code, sample_rate_hz	u8, u16	as recorded
first_sample, last_sample	u32	the DRDY counter at the ends of the file
frames, gaps, signature_blocks	u32	counted as written
public_key_fingerprint	string	the section 7.4 form, so the card can be verified without the unit
sha256	string	over the .eeg file, written last

Rules that follow from the above and are part of the definition. A session file whose sidecar is missing – power lost mid-session – **is still a valid recording**: the reader rebuilds every sidecar field except utc_start and sha256 from the frames themselves, and records the sidecar as absent rather than the session as failed. A file is never appended to after STOP_SESSION, and a <session_id> is never reused. Card-full policy is **stop and flag in STATUS, never overwrite**; the free-space field of section 5.4 is what the host watches. At the 50.7 kB/s of section 5.8 a recording hour is **182.5 MB**, a three-hour session **548 MB** and a three-session loan about **1.10 GB** (calculated), so a 32 GB card holds roughly 175 recording hours and never fills within a loan.

Nothing above has been written by a real firmware. *Corrected 2026-09-02: this paragraph said sd_append() and sd_free_mb() “are stubs”. They are not stubs any more – they are written, at drivers.c:163 and drivers.c:175 (section 1.2).* What has not changed is the part that matters: **no card has ever been written**, because no unit exists, and the SDMMC host did not initialise under QEMU because QEMU has no card slot. Nothing writes the sidecar. This is still the contract the driver must meet, not a description of an existing card.

6. The command channel

6.1 Host to device

A command is a normal frame with type = 16 (CMD), first_sample = 0, rate_code = 0, n_samples = payload length. The payload is the opcode byte followed by its arguments. This is what provision.py and the browser tool both send.

rx_poll() now reads it that way (FW-D14, fixed 2 September 2026): after the CRC check it requires at least FRAME_HDR_BYTES + 1 bytes, requires out[0] == PROTO_VERSION and out[1] == FT_CMD, and dispatches from out + FRAME_HDR_BYTES. A frame that fails any of those is dropped in silence, which is the right answer on a link where the alternative is acting on a frame the device did not understand.

Until that fix the whole decoded body went to handle_command(), which read byte 0 – the protocol version – as the opcode. PROTO_VERSION is 1 and CMD_START_SESSION is 0x01, so every command from the host started a recording session, including the two that exist precisely so a host can talk to a unit without starting one. The lesson is in

section 8: a simulated device written from the same specification as the firmware shares the firmware's misreadings of it, and will not find this class of defect.

Four images, not five, and the name has an underscore in it. This command and ASM-EEG-007 section 6.1 step 4 used to disagree twice, and a factory can only follow one of them. Both are settled against the source rather than by preference:

- The image is `eeg_field_kit.bin`. `firmware/CMakeLists.txt` line 7 is `project(eeg_field_kit VERSION 0.2.0)` and ESP-IDF names the binary after the project, so ASM-EEG-007's `eeg_fieldkit.bin` was a file that would never exist. Corrected there.
- `0x930000 storage.fat` is REMOVED from this command. There is no `storage.fat` anywhere in the package to flash, and nothing in the firmware mounts the internal storage partition: the only FAT mount is `esp_vfs_fat_sdmmc_mount()` in `drivers.c`, which mounts the microSD CARD. The partition is allocated in `partitions.csv` and is currently unused. Leaving the line in would have stopped the flash step with a missing file, on every unit, at the factory.

6.2 Acknowledgement

Every command produces one `CMD_ACK` frame (type = 6) whose payload is:

Offset in payload	Size	Field
0	1	opcode echoed
1	1	reserved, zero
2	1	status code
3	1	result length, bytes following
4	n	result bytes

In absolute frame offsets that puts the status byte at 12 and the result at 14, which is what `provision.py` reads.

`main.c` sends this shape as of 2 September 2026. `ack_emit()` (`main.c:430-439`) is the single place a `CMD_ACK` is built for the general command channel, and the browser tool's own `parseAck()` decodes it in the interop harness of section 8. Before that, every ack went out as `{opcode, status}` with the status at offset 1 and no length, and IDENTIFY, LOOPBACK and CLOCK_XCHG skipped even that and returned their result at offset 0 with no opcode echo at all – so a host could not tell which command an acknowledgement answered and matched replies by arrival order.

Every path uses it now. *Corrected 2026-09-02: this paragraph read “One path still does not use it”, and named `handle_provision()`.* The whole `0x40-0x4F` family returns through `ack_emit()` at `main.c:561`, including the refusal a command gets when it arrives outside provisioning mode, so FW-D20 is closed. It mattered: `provision.py` reads the status at absolute frame offset 12, which is payload offset 2, and under the old shape that was the first *result* byte – on `0x48 READ_PROVISION_STATE`, the config-zone lock flag. An **unlocked** unit therefore reported 0 and the tool read it as success. Three of the interop harness's 57 checks now read a provisioning status at this offset, so

the shape cannot drift again without a test failing. End-of-line provisioning has still never been run against a real ATECC608B.

Status	Meaning
0x00	OK
0x01	unknown opcode
0x02	bad length
0x03	argument out of range
0x04	wrong state (for example a block command outside a session)
0x05	interlock: VBUS_DET high, session refused (S-01)
0x06	not provisioned
0x07	hardware fault (bus NAK, converter not responding)
0x08	payload CRC or hash mismatch
0x09	locked: the ATECC configuration zone is already locked
0x0A	timeout
0x0B	not implemented in this build – the honest answer from a stub

The source uses 0x02 for two different things, and this table is only half of what a host sees. Recorded 2026-09-02. CMD_ENTER_PROV (0x40) refuses when the ATECC608B configuration zone is already locked – which is the guard that stops a fielded unit being re-keyed over USB, and is intended – and it answers **0x02** for it, where this table assigns 0x02 to “bad length” and 0x09 to “locked”. CMD_ATECC_WRITE_CONFIG (0x4B), in the same handler, answers 0x09 for the same condition. So an operator reading this table alone, faced with a 0x02 from step 1, would go looking for a framing fault that is not there.

The code is the fact and the document does not get to overrule it, so it is written down rather than quietly renumbered: **0x02 from 0x40 means “already provisioned, refused”, and 0x02 from anything else means bad length.** provision.py hints both readings at that step. The fix is one line in main.c – return 0x09 there, which is defined for exactly this and is what 0x4B already returns – and it belongs to whoever owns main.c; it is carried in section 10 rather than pre-announced here as done.

handle_provision() also returns several codes this table does not define at all: 0x10 to 0x18 for individual driver failures (GenKey, pubkey read, serial read, each NVS write, the config write, the lock) and 0x20 for a short payload on 0x44 and 0x45. They are distinguishable and they are useful at a station; they are not in section 6.2 and a host written to this table will render them as unknown. Also carried in section 10.

Any command that receives no acknowledgement within **2 s** is a failure; the host retries once and then reports. Long operations (CMD_TIMING_SELFTEST, CMD_ATECC_GENKEY) are given **20 s**.

6.3 Command table

Opcode	Command	Arguments	Result	Errors	Test
0x01	START_SESSION	session_id 16 B	–	0x04, 0x05 if VBUS_DET high	T21
0x02	STOP_SESSION	–	–	0x04	T21
0x03	BLOCK_START	label_len u8, label <= 32 B	–	0x02, 0x04	T13
0x04	BLOCK_END	–	–	0x04	T13
0x05	SET_RATE	rate_code u8	–	0x03	T7, T14
0x06	SET_GAIN	channel u8 1-16, gain_code u8 0-6	–	0x03	T7
0x07	IMPEDANCE	mode u8 (0 off, 1 AC 7.8 Hz, 2 AC 31.2 Hz), mask u16	–	0x03, 0x07	T10
0x08	RETRANSMIT	seq_from u16, seq_to u16	frames replayed u16	0x03 if outside the ring	T15
0x09	TIMING_SELFTEST	n_tones u8 (40)	median x100 i16, p95 x100 i16, verdict u8, 40 residuals i16	0x0B until the codec driver has been brought up on hardware; main.c returns a 3-byte result today, not the result column above	T13

Opcode	Command	Arguments	Result	Errors	Test
0x0A	LIGHTS	mode u8 (0 off, 1 auto, 2 green, 3 red, 4 amber), optional mask u8	–	0x00 for modes 0 to 4; 0x0B UNIMPLEMENTED for any other mode; 0x02 BAD_LENGTH if the mode byte is absent. <i>Corrected twice on 2026-09-02. First: the phase driver now exists (section 3.2), so mode 1 shows all three colours. Then FW-D18: this row said modes 2, 3 and 4 were not implemented and that the source wrongly answered 0x00 OK to them. They are implemented now – the mask in c[2], when present, selects the sites the forced colour applies to, and defaults to all eight. A host that asks for a mode the firmware does not have gets a refusal rather than a false success</i>	T11

Opcode	Command	Arguments	Result	Errors	Test
0x0B	CLOCK_XCHG	host time i64 us	sample_index u32, device time i64 us	–	–
0x0C	PLAY_AT	asset_id u8, start_sample u32, level i16 tenths dB	–	0x04, 0x0B	T12b, T13
0x0D	FW_UPDATE_BEGIN	image_len u32, sha256 32 B	slot u8	0x04 during a session	T25
0x0E	FW_UPDATE_DATA	offset u32, bytes ≤ 512	bytes accepted u16	0x08	T25
0x0F	IDENTIFY	–	proto version u8, fw major u8, fw minor u8, board revision letter u8, ring bytes u32, capability flags u32, current rate code u8, supported rate count u8, unit serial NUL-terminated	–	TOOL-EEG-022 step D1
0x10	LOOPBACK	up to 240 bytes	the same bytes, unchanged	none: the source truncates a longer payload to 240 bytes rather than refusing it	TOOL-EEG-022 steps D4-D6
0x11	SET_HP_LEVEL	level i16 tenths dB	level read back from the codec	0x03, 0x0B	T17
0x12	REF_TONE	freq u16 Hz, level i16, duration u16 ms	env_voice u16, env_room u16	0x0B	T17, E-16
0x13	LOAD_AUDIO_BEGIN	asset_id u8, len u32, sha256 32 B	–	0x03	T12b

Opcode	Command	Arguments	Result	Errors	Test
0x14	LOAD_AUD IO_DATA	offset u32, bytes <= 512	bytes accepted u16	0x08	T12b
0x15	LOAD_AUD IO_END	sha256 32 B	–	0x08	T12b
0x16	MIC_MUTE	state u8	state read back	–	T17
0x17	GET_VERS ION	–	version 3 B, image sha256 32 B, board rev 1 B	–	T25
0x18	FW_UPDAT E_END	sha256 32 B	–	0x08	T25
0x40- 0x49, 0x4F	provisioning block	section 7.3	section 7.3	0x06, 0x09	T6
0x4A	READ_CALI BRATION	offset u16 LE, length u8	that slice of the stored calibration blob	0x02 short payload, 0x06 nothing stored	T6
0x4B	ATECC_W RITE_CON FIG	block u8, mask 32 B, image 32 B (65 B with the opcode)	–	0x02 short payload, 0x09 zone already locked	T6

Added 2026-09-02. The two rows above were missing. Both opcodes exist in `main.c` and both are sent by `provision.py`, so a table that stopped at 0x49 described neither the firmware nor the tool. 0x4A `READ_CALIBRATION` is what makes TST-EEG-004 **T6's acceptance limit executable at all** – “constants read back byte-identical to those written” – because until it existed nothing in the package could read the calibration blob back. It is chunked because a full calibration set does not fit in one frame, and it is **exempt from the provisioning-mode gate**, because T6 runs after provisioning on a unit that has left that mode and may have had its zone locked. 0x4B `ATECC_WRITE_CONFIG` writes one masked 32-byte block of the ATECC608B configuration zone; a word with an all-zero mask keeps its factory value.

These two collided once and the collision is worth remembering. On 2 September both `main.c` and `provision.py` allocated 0x4A independently – the firmware to the calibration reader, the tool to the config write – because each took the next free opcode without the other. Had it shipped, step 7b would have sent a 3-byte read to a handler expecting 65 bytes and T6's read-back would have failed on every unit. `main.c` is the authority, `provision.py` moved its config write to 0x4B, and `provision_selftest.py` now asserts that every `CMD_` constant in the tool is unique, which is the check whose absence allowed it. The interop harness carries the same guard from the other side: one of its 49 checks is “0x4A is still the calibration reader, not the config write”.

SET_HP_LEVEL must clamp the codec volume register at the value recorded at calibration. RFQ E-29 caps the headphone output at 100 dB SPL at any commanded level, and the calculated full-scale output of the fitted codec and a 47 Ohm load is about 110 dB SPL, so the clamp is the only thing that meets the requirement. The clamp is **not implemented** today; the codec driver itself is a stub.

Two corrections to this table, 2 September 2026. 0x0F was FW_UPDATE_END here and CMD_IDENTIFY in main.c and in TOOL-EEG-022, and there was no 0x10 row at all against their CMD_LOOPBACK. A specification that collides with the source and with the tool written to it is worse than no specification, because each side can be checked against it and pass. The two connectivity commands keep the opcodes the source and the tool already use, and **FW_UPDATE_END moves from 0x0F to 0x18**, the first free opcode above the audio group. Nothing implements the firmware-update family yet, so the move costs nothing; moving IDENTIFY instead would have broken a shipped tool. 0x0D FW_UPDATE_BEGIN and 0x0E FW_UPDATE_DATA keep their opcodes, which is what main.c's CMD_FW_UPDATE_BEGIN 0x0D already says.

main.c's CMD_PROVISION 0x0E is **retired**; 0x0E is FW_UPDATE_DATA and provisioning mode is entered with 0x40. The enum in main.c still carries the retired name and it should be deleted (FW-D10).

The opcodes main.c handles today are **0x01-0x05, 0x08-0x0B, 0x0F and 0x10**, eleven in all, plus the **thirteen** provisioning opcodes 0x40-0x4B and 0x4F, which handle_provision() implements in full. *Corrected 2026-09-02: this sentence said “the eleven provisioning opcodes 0x40-0x49 and 0x4F”, which predates 0x4A and 0x4B.* The rest are FW-D10.

CMD_IDENTIFY's capability flags are bit 0 CDC, 1 WebUSB, 2 microSD mounted, 3 codec initialised, 4 ATECC608B present, 5 provisioned. Bits 3, 4 and 5 were wrong until 2 September: nothing ever set bit 3, so a working codec always read as absent, and bit 4 was set only when the configuration zone was locked, which answers “has this unit been provisioned” when a production tester is asking “is the part fitted”. They are now drv_codec_ready(), drv_atecc_present() and drv_atecc_config_locked() respectively – three different questions with three different answers. TOOL-EEG-022 section 2.3 defines the same six bits and agrees.

7. End-of-line provisioning

This is the step that turns an assembled board into a serialised instrument. It is TST-EEG-004 **T6** and it runs **after** the characterisation steps T7, T10, T12 and T17, because it writes their results into the unit. Only T5a precedes it.

7.1 The station

One PC (Windows 11 or Ubuntu 22.04), Python 3.11, pyserial, cryptography.
Offline. An ESD bench, a USB hub, and a barcode scanner for the board serial. No signing key and no network credential is present on this station or in the building (F-19). The unit is powered from its own battery.

Two different cables reach the unit at this station and they must not be confused. The firmware image goes in through the **DevKitC-1's own UART USB-C port**, reached through the MP-01 opening, because that port carries the auto-reset circuit (section 2.3). The provisioning session that follows runs over the instrument's own USB link, through the ADuM4160 module's host USB-C, which is what a participant will later use.

7.2 The command

```
python3 tools/provision.py --port /dev/ttyACM0 --serial TI0V-B-0007 \  
    --calibration cal_TI0V-B-0007.json \  
    --out records/TI0V-B-0007.json
```

The serial passed to `--serial` is a TI0V-B-nnnn string. **The format is defined once, in PKG-EEG-015 section 5, and this document cites it rather than restating it**; what belongs here is only what the script does with it. `provision.py` rejects any argument that does not match the registered format, writes the string unchanged into the USB `iSerialNumber`, the ATECC608B data zone and the provisioning record, and uses it as the `<unit_serial>` directory name of section 5.9, so the label, the descriptor, the card and the record cannot disagree.

`--dry-run` prints the whole sequence without touching a device, so the operator can be trained and the station validated before a board exists. `STATION` and `OPERATOR` environment variables are recorded into the output.

7.3 The steps

Rewritten 2026-09-02, from the script. This section used to be headed “The eleven steps” and claimed: “this table is the script’s step list, read out of `firmware/tools/provision.py`, and where the two ever differ the script is what runs.” The second half of that sentence was right and the first half was not. The table was **not** the script’s step list. It had no configuration-zone write at all, and it put the lock at step 8, after GenKey. The script writes and locks the zone at **2b** and **2c**, before GenKey.

The order in the old table could not have worked, and following it would have scrapped parts. A factory-fresh ATECC608B will not generate a key into a slot whose configuration has not been written and locked, so step 3 fails on every unit; and step 8 would then lock whatever the part arrived with – Microchip’s default configuration – which is irreversible, destroys the breakout and produces no device identity. The write and the lock therefore run first. Old step 8 becomes the read-back that proves the lock took, which is what TST-EEG-004 T6’s acceptance limit actually asks for; old step 9 read the same state with the same `0x48`, so the two are printed as one step, 8/9, rather than sending an identical command twice with nothing in between. A new step **7b** reads the calibration constants back and compares them, which is the other half of that limit.

Each step prints its name, then `ok`, `FAILED: <reason>` or `skipped (dry run)`. A failure writes the partial record and exits with status 1. Nothing is retried automatically, because two of these steps are irreversible. The step numbers below are the script’s own, including its lettered ones; where this table and the script ever differ again, **the script is what runs**.

#	Step	Opcode	Writes	Irreversible	Gate
1	enter provisioning mode	0x40	–	no	refused with 0x02 if the zone is already locked – see section 6.2
2	read the ATECC608B serial number	0x43	–	no	

#	Step	Opcode	Writes	Irreversible	Gate
2b	write the configuration zone from the template	0x4B, once per 32-byte block whose mask is non-zero	the ATECC608B config zone, masked	no	only with --write-config
–	<i>stop</i>	–	–	–	if --lock was given and 2b did not write, the run ends here rather than locking a zone the station never wrote
2c	lock the ATECC608B configuration zone	0x47	the config zone	yes	only with --lock, and only after the serial is typed back or passed with --confirm-serial
3	generate the P-256 key pair in slot 0	0x41	the private key, inside the element	yes	20 s deadline
4	read back the public key	0x42	–	no	20 s deadline
5	write the USB vendor and product identifiers	0x44	NVS: usb_ids, vid u16 + pid u16	no	
6	write the hardware revision	0x45	NVS: hw_rev	no	
6b	write the unit serial into the device	0x49	NVS: unit_serial, the TI0V-B-nnnn string	no	
7	write the calibration constants	0x46	NVS: calib, the blob from --calibration	no	skipped with no --calibration

#	Step	Opcode	Writes	Irreversible	Gate
7b	read the calibration constants back and compare	0x4A, chunked	—	no	this is TST-EEG-004 T6's acceptance limit: byte-identical or the step fails
8/9	read the state back and confirm the lock	0x48	—	no	if --lock was given and the zone reads unlocked, the operator is told to hold the unit and report it
10	leave provisioning mode	0x4F	reboot; iSerialNumber is now the TIOV-B-nnnn written at 6b	no	

Both irreversible steps are behind flags. Neither --write-config nor --lock is implied. Without --lock nothing in a run is irreversible and the run cannot produce a finished unit, which is what a training or station-validation run wants. --serial is checked against the PKG-EEG-015 section 5 format before anything is sent.

The private key is generated **inside** the secure element and cannot be read out by anyone, including the programme. Step 2c cannot be undone: a config zone locked with the wrong template scraps the ATECC breakout. J11 is socketed, so the loss is one breakout and not one board, and the manufacturer is required to hold 10 % spare ATECC608B breakouts.

Where this document and the script disagree about NVS, the firmware is the fact. The script's own header still says the identity and the constants land in namespace "tioV" in the **default** nvs partition, and warns that drivers.c erases that partition unconditionally on ESP_ERR_NVS_NO_FREE_PAGES. That was true and **is not true any more**: drivers.c now routes the calib key to partition calib / namespace eegcal and every identity key to prov / eegcfg, initialising each on first use, which is what section 2.5 specifies. The stale warning is in provision.py and in README_provisioning.md, neither of which this document owns; it is raised in section 10.

The whole of section 7 is still unexecuted. No step above has ever been sent to a real ATECC608B. The interop harness drives 0x4A, 0x4B and the mode gate against a simulated part, which proves the framing and the refusals and nothing about the silicon.

The ATECC608B-TNGTLS variant must never be substituted. It arrives with its config zone already locked and its keys owned by Microchip; F-18 and E-21 are impossible on it. The AVL alternate column must say so.

No eFuse is burned during provisioning on Phase 1. Section 7.5 says when eFuses are burned and when they are not.

7.4 What the script prints and what it returns

On success it prints the unit serial, the ATECC serial and the public-key fingerprint, then the instruction to print the fingerprint on the enclosure label (M-03) and return the record with the unit. The record is records/<serial>.json:

```
{
  "unit_serial": "TIOV-B-0007",
  "hardware_revision": "B",
  "board": "EEG-CAR-01 Rev B",
  "provisioned_utc": "2026-09-01T09:14:02Z",
  "station": "...", "operator": "...",
  "usb_vid": "0x1209", "usb_pid": "0x0001",
  "atecc_serial": "0123...",
  "device_public_key": "<128 hex chars, raw uncompressed X||Y>",
  "public_key_fingerprint": "A1B2 C3D4 E5F6 0718",
  "provision_state": "...",
  "calibration": { ... },
  "steps": [ {"step": "...", "result": "pass"}, ... ]
}
```

The public-key fingerprint is defined here and nowhere else. It is the **first 8 bytes of SHA-256 over the 64-byte uncompressed public key**, rendered as **16 uppercase hexadecimal characters in four groups of four separated by single spaces**, for example A1B2 C3D4 E5F6 0718. ASM-EEG-007, QP-EEG-010, PKG-EEG-015, TST-EEG-004 and REG-EEG-012 cite this definition and do not restate it. The fingerprint is what goes on the M-03 label, and the label is generated from this record so the two cannot disagree. All records for a shipment are returned as one manifest.

7.5 How the signing key never leaves the programme, and when eFuses are burned

Who	Holds	Does
Programme, Brussels	the RSA-3072 secure-boot private key, offline, two custodians, two backups	builds, signs with espsecure.py sign_data, publishes the release
Manufacturer	pre-signed binaries and the public-key digest file only	flashes, provisions, tests, and from Phase 2 burns eFuses

Phase decision, written down and not deferred: no eFuses are burned on the two Phase 1 prototypes. They ship with secure boot and flash encryption disabled and run unsigned images, so the firmware volunteer can re-flash them freely during bring-up. This is a scoped deviation from F-19 for two units that are never given to a participant. **TST-EEG-004 T25 is therefore a Phase 2-onward step and is not a Phase 1 gate**; it is not run on the prototypes and its absence is not a Phase 1 non-conformance.

From Phase 2 onward the burn happens at the station, in this order, each step irreversible:

1. espfuse.py burn_key_digest secure_boot_digest.bin BLOCK_KEY0 SECURE_BOOT_DIGEST0
2. espfuse.py burn_efuse SECURE_BOOT_EN 1
3. first boot: the bootloader generates the flash-encryption key **on the device**, so the manufacturer never holds that either
4. espfuse.py burn_efuse DIS_DOWNLOAD_MANUAL_ENCRYPT SOFT_DIS_JTAG

5. espfuse.py summary captured verbatim into the per-unit record

TST-EEG-004 places that burn after T5 to T17 and before T18, and requires one deliberate negative test per batch: attempt to boot an unsigned image and confirm rejection (T25).

Once burned, the ESP32-S3 module **is** part of the instrument identity, together with the ATECC608B at J11. It is not a field-swappable part, notwithstanding the general module philosophy of DSN-EEG-003 section 2. A failed MCU module means the unit returns to the programme for re-provisioning under a new record. On the Phase 1 prototypes, where nothing is burned, the module remains swappable.

8. The host verification tool

firmware/tools/verify_stream.py is the decoder the production test uses, and the same tool the programme uses to read a returned microSD card, laid out as section 5.9 defines. It reads from a file or a serial port, and reports on the four failures that are otherwise invisible.

```
python3 verify_stream.py --file session.bin --pubkey records/TI0V-B-0007.json
python3 verify_stream.py --port /dev/ttyACM0 --seconds 60 --pubkey ...
--json t14.json
```

Check	How
corrupted frame	CRC-32 over the decoded body, counted as BAD_CRC
lost frame	sequence continuity modulo 65536, and GAP frames
discontinuous timeline	first_sample continuity against n_samples, resuming after a GAP
unverifiable block	ECDSA P-256 over the chained digest, against the exported public key

It prints one JSON report and exits **0** on pass, **2** on fail, so it can be a test step directly:

```
{
  "frames": {"DATA": 90000, "STATUS": 1800, "SIGNATURE": 43},
  "samples": 1800000, "sample_rate_hz": 1000, "duration_s": 1800.0,
  "first_sample_index": 41, "last_sample_index": 1800041,
  "gap_frames": 0, "samples_lost_to_gaps": 0,
  "signature_blocks": 43, "signature_blocks_verified": 43,
  "errors": [], "error_count": 0, "pass": true
}
```

Step	Use	Pass condition
T5	--port --seconds 10 on each of Windows 11, macOS and Linux, after the OS has enumerated the device	frames decode with zero bad CRC on every host. The descriptor dump itself comes from pnutil /enum-devices, ioreg -p IOUSB and lsusb -v, not from this tool
T13	--port capturing while CMD_TIMING_SELFTEST runs	zero CRC or continuity errors across the 40 tones; the median and p95 come from the CMD_ACK
T14	--file on the 30-minute host capture and --file on the card copy, then compare the two reports	DATA = 90,000, error_count = 0, and the two reports identical
T16	--file --pubkey records/<serial>.json over a session of at least 8192 samples	signature_blocks_verified = signature_blocks, and signature_blocks > 0

That last condition has to be asserted separately by the operator: the tool's pass flag is true when `blocks == 0`, so a stream carrying no SIGNATURE frames at all would pass. This is stated rather than hidden, and the fix is listed in section 10.

The T-numbers above are TST-EEG-004 Rev C's and are cited, not coined. Where a step this document needs does not exist there, it is raised as an open item rather than numbered here.

8.1 The protocol interoperability harness

`webtest/tests/interop/` is a second host-side tool, added on 2 September 2026, and it answers one question: **do the firmware and the browser test program speak the same protocol?** Run it with `sh webtest/tests/interop/run.sh`. It needs a C compiler and Node 18 or later, and it needs neither ESP-IDF nor hardware.

It compiles the shipped `firmware/main/main.c` against small ESP-IDF stubs, simulates the peripherals in `drivers_sim.c`, wires the two USB endpoints to `stdin` and `stdout`, and drives the result with `webtest/js/protocol.js` – the same module the browser tool ships. It reports **57 checks, all passing** as this document is issued. *Corrected 2026-09-02: this paragraph said 32, which was the count before the provisioning-acknowledgement and opcode work of that day.* The checks are frame decode and CRC, IDENTIFY parsed by the host's own decoder including all six capability bits and the 6 MiB ring of FW-D13, LOOPBACK at four payload lengths, opcode echo on seven commands and the twelve-byte CLOCK_XCHG result, both halves of the S-01 interlock, silence in reply to a corrupted CRC and to random noise, and – new on 2 September – the provisioning family: a config write refused outside provisioning mode, opcode echo on 0x4B, a well-formed config write accepted after 0x40, a short config write refused with bad length, and **“0x4A is still the calibration reader, not the config write”**, which is the guard against the opcode collision described in section 6.3 recurring.

It exists because the test that came before it could not have found FW-D14. That test drove the real host code against a device **simulated in JavaScript**, and the simulation was written by hand from this specification, as `main.c` was. The two drifted from it independently and in different directions – `main.c` dispatched on the frame's first byte, the simulation answered `{opcode, 0xFF}`, and section 6.2 specified a third shape – and every test passed throughout, because the JavaScript host and the JavaScript device shared one misunderstanding. A simulator written from the same document as the firmware tests the pair of readings, not the document.

What it does **not** do: it says nothing about the analogue front end, the converters, the timing, the descriptors (the stubs expand the TinyUSB macros to placeholders, and `run.sh` neutralises the `_Static_assert` of FW-D11 in its private copy), or anything else that needs a board. It retires no row of section 1.3 that needs a unit.

Corrected 2026-09-02. This paragraph used to end “and FW-D20 is a defect it looks straight at and does not see.” That was true of the 32-check version, which checked only that 0x48 echoed its own opcode – a test the wrong ack shape passes. The provisioning checks added on 2 September read a **status** at the section 6.2 offset, and one of them expects a specific 0x02 bad-length refusal, so the shape is now covered from the host side. It is worth keeping the old sentence visible: a harness can look straight at a defect and pass, and the fix was to make it read the field the defect was in.

8.2 The QEMU run, and exactly what it proves

New, 2 September 2026. The firmware has been **run**, once, under `qemu-system-xtensa 9.0.0 (esp-develop)` with `-M esp32s3 -m 4M`, against a `sdkconfig.defaults + sdkconfig.phase1 + sdkconfig.qemu` build. The complete console capture is `firmware/release/qemu_boot.log` and it is part of the release.

This is an emulator, not a unit, and the distinction is the whole of what follows. QEMU's esp32s3 machine has no octal PSRAM, no microSD, no ES8388 and no ADS1299. Every peripheral this instrument exists to talk to is absent.

What the run **does** prove – the part that needs no peripheral:

Observed in the log	What it settles
ROM loader, then ESP-IDF v5.2.5 2nd stage bootloader	the built bootloader is a bootloader and this is the version that built it
the nine-row partition dump, matching <code>partitions.csv</code> offset for offset	the flashed table is this table, and <code>nvs</code> , <code>otadata</code> , <code>phy_init</code> , <code>factory</code> , <code>ota_0</code> , <code>ota_1</code> , <code>calib</code> , <code>prov</code> and <code>storage</code> all land where section 2.5 says
Defaulting to factory image, five segments loaded, Loaded app from partition at offset 0x20000	the image is loadable from the factory slot at the section 9 offset, and its segment map is consistent
Project name: <code>eeg_field_kit</code> , App version: <code>e91f9d58-dirty</code>	the application descriptor carries the CMake project name of section 9 – and a dirty tree, which is why section 2.1 says the build is not yet reproducible

Observed in the log	What it settles
main_task: Calling app_main() then the firmware's own first log line	app_main() is reached and runs
drv_sd_init() and drv_codec_init() both time out, log, and continue	the two degrade-gracefully paths behave as written: a unit with no card or no codec does not refuse to start
the FW-D13 diagnostic, verbatim, then abort()	the ring-buffer guard added on 2 September fires and names the cause, rather than asserting somewhere inside xRingbufferSend a frame later

What it proves **nothing** about: every register value, the daisy-chain order, the SPI timing, the ADS1299 entirely, the USB descriptors and enumeration, the contact lights, the ATECC608B, the fuel gauge, timing accuracy, and anything else in TST-EEG-004. No row of section 1.3 is retired by it and no requirement is met by it.

The abort at the end is the correct behaviour, not a failure of the run. PSRAM is absent, so the 6 MiB ring of F-06 cannot be allocated, and app_main() refuses to carry on without it. sdkconfig.qemu sets CONFIG_SPIRAM_IGNORE_NOTFOUND=y only so the boot gets far enough to reach that point instead of failing earlier in cpu_start; it is not a configuration for any unit (section 2.4).

One thing the log raises that is not a QEMU artefact. A unit that boot-loops or aborts in a participant's home cannot be told apart from a dead battery or a bad cable, and the browser tool cannot ask it what is wrong because it never enumerates. The abort is right for a bench and wrong for a field device, and what a field device should do instead – enumerate, report the fault in STATUS, and refuse to record – is not written. That is carried in section 10.

9. Firmware release process

Version scheme. FW-EEG-001 v<major>.<minor>.<patch>, carried in the CMake project(... VERSION ...) field, in the ESP-IDF application descriptor, in the STATUS frame and in CMD_GET_VERSION. The shipped project declares **0.2.0**. v0.x is Phase 1 bring-up; v1.0.0 is the first fleet release and is gated on the safety review sign-off, which has not happened.

Reproducible build. FROM espressif/idf:v5.2.5 pinned by image digest, SOURCE_DATE_EPOCH set from the release commit date, one make release. The manufacturer is invited, not required, to reproduce the hash. **This has not yet been done.** The images described below were built on a developer machine from a working tree with uncommitted changes – the boot log records the application version as e91f9d58-dirty – so the SHA-256 list below is the hash of *those* images and not yet a hash anyone else can reproduce. Rebuilding from a clean tree in the pinned container is an open item.

Release contents, flash offsets and what was actually built. *Measured 2026-09-02.* The four files below are in firmware/release/ with a manifest.json carrying their SHA-256; the figures are idf.py size and the files on disk.

File	Offset	Bytes	SHA-256 (first 16 hex)
bootloader.bin	0x0	23,168	23864fcd3a087b98
partition-table.bin	0x8000	3,072	fdd6a7170583bbaf
ota_data_initial.bin	0xF000	8,192	7d2c7ac4888bfd75
eeg_field_kit.bin	0x20000 (the factory slot)	405,360	90e5ec1f6b91fb4d

manifest.json carries the full 64-character digests and is the file the manufacturer verifies against; the truncations above are for reading, not for checking.

Memory, and the one figure that is a problem. *Added 2026-09-02; the whole of this paragraph is new, and item 17 of section 10 is the open half of it.*

Figure	Value	Reading
Total image, as idf.py size reports it	405,245 bytes	13 % of the 3 MB factory slot. Ample
eeg_field_kit.bin on disk	405,360 bytes	the same image plus the padding and appended SHA-256 that esptool writes into the flashable file
Flash code / rodata	237,099 / 75,468 bytes	
DIRAM used	88,799 of 345,856 bytes (25.7 %)	257,057 bytes remain. Ample
Static IRAM used	16,383 of 16,384 bytes – one byte free	not a pass. See below

The IRAM figure is the finding. One byte free is a cliff, not a margin: the next function anyone marks IRAM_ATTR fails the link with an error naming a section rather than a cause, and this design has more interrupt work coming – E-13's tone scheduler and E-12's onset detector are both interrupt-side work that does not exist yet.

Two ISRs that this firmware does not use were taken out of IRAM to make room – CONFIG_SPI_SLAVE_ISR_IN_IRAM=n and CONFIG_GPTIMER_ISR_HANDLER_IN_IRAM=n; there is no SPI slave anywhere in the design and nothing uses gptimer. **It did not help.** The figure came back 16,383 of 16,384, byte for byte. The change is kept, because carrying interrupt handlers for a bus and a timer this firmware does not use is wrong either way, but **it is not the fix and must not be recorded as one.** SPI master stays in IRAM deliberately: sample_task() reads the converters on every DRDY and that path must survive a flash-cache stall.

What is not yet known is whether the pool is genuinely full or whether esp_idf_size is reporting against a fixed 16 kB window that is not the real limit on an ESP32-S3 with octal SPIRAM and XIP. Settling that needs the linker map read against hardware. It is

carried as section 10 item 17 and as SIM-EEG-018 open item 1, and it is **not** guessed at from here.

storage.fat was a fifth row here and is withdrawn: no such file exists in the package and nothing in the firmware mounts the internal storage partition it would have been written to. See the note in section 6.1.

```
esptool.py --chip esp32s3 --port <devkit UART port> --baud 921600
write_flash \
  --flash_mode qio --flash_size 16MB \
  0x0 bootloader.bin 0x8000 partition-table.bin 0xF000
ota_data_initial.bin \
  0x20000 eeg_field_kit.bin
```

<devkit UART port> is the DevKitC-1's own UART USB-C port, per section 2.3. esptool.py drives the DevKit's auto-reset circuit itself; there is no separate boot-mode step and no fixture on J26.

manifest.json accompanies every release: version, ESP-IDF commit, esp_tinyusb version, build container digest, SHA-256 of every file, the WinUSB GUID, the board_pins.h regeneration record, and the secure-boot public key digest. The manufacturer verifies the SHA-256 list before flashing and records **which version and which image hash went onto which unit** into that unit's record. A circulating kit that visits twenty participants will be re-flashed several times over its life; without that record nobody can say what a given recording was produced by, and a pre-registered study cannot tolerate that.

A/B update with rollback (F-20). ota_0 and ota_1 alternate; factory is the recovery image, flashed once at end of line and never overwritten in the field. A new image arrives over CMD_FW_UPDATE_BEGIN / DATA / END, is written to the inactive slot, and boots in PENDING_VERIFY. esp_ota_mark_app_valid_cancel_rollback() is called **only** after three self-checks pass: the device enumerated on USB, the ATECC608B serial was read, and one ADS1299 RDATA read returned. If any check fails or the unit resets first, the bootloader reverts to the previous slot on its own. Anti-rollback is on from Phase 2, so a downgrade below the burned secure version is refused; on the Phase 1 prototypes, which burn no eFuses, there is no anti-rollback and a downgrade is possible.

The manufacturer never builds firmware and never modifies source. Any change goes through the programme and produces a new version.

10. Open items, honestly

#	Item	Consequence	Owner	Gate
1	The firmware has never been compiled against a real ESP-IDF installation. CLOSED 2026-09-02. It was built against ESP-IDF v5.2.5, clean at ESP-IDF's default -Wall -Werror=all; images in firmware/release/. Five project defects surfaced in the process and are recorded in sections 2.2 and 2.4	Every ESP-IDF header, Kconfig option and TinyUSB macro this source names now resolves, and both _Static_asserts held. It says nothing about whether any of it is <i>correct</i>	firmware volunteer	closed. What replaces it is items 1a and 2
1a	The build is not reproducible yet. It was made on a developer machine from a dirty tree, not in the pinned espressif/idf:v5.2.5 container from a release commit	The SHA-256 list in section 9 cannot be independently reproduced, and F-19's "the manufacturer verifies before flashing" is a weaker guarantee than it reads	programme	a clean-tree container build whose hashes match
2	It has never run on hardware. It has run once under QEMU (section 8.2), which is a much weaker claim: QEMU's esp32s3 has no PSRAM, no microSD, no ES8388 and no ADS1299, so no peripheral this instrument uses was present	Every register value, the daisy order, the SPI timing and the descriptor set are assumptions. The QEMU run retires no row of section 1.3	firmware volunteer	T5, then T13

#	Item	Consequence	Owner	Gate
3	The five drivers are written and none has run. <code>drivers.c</code> implements them (section 1.2); no I2C address, register write or timeout in it has ever been answered by a part. The block-signing task is still not written and not declared	TST-EEG-004 steps T12, T13, T14, T16 and T17 cannot be signed off on code inspection, and T16 has no input at all while nothing emits SIGNATURE frames. The total step count is TST-EEG-004 Rev C's to state, not this document's	firmware volunteer	each driver has a named step
4	The contact-light phase driver does not exist CLOSED 2026-09-02 (section 3.2, FW-D16). <code>lights_phase()</code> and <code>lights_task()</code> drive LED_V against the shift register in antiphase, and the three colours come from both halves of the lead-off measurement. E-27 is met in the source. The alternation is about 250 Hz, not the nominal 240, because it quantises to the 1 kHz tick; the requirement is "above 100 Hz"	Written, never run. T11 is no longer blocked by a missing driver; it still needs a unit and a colorimeter. What remains open is narrower: CMD_LIGHTS modes 2, 3 and 4 – forcing a colour – are not implemented and the source answers 0x00 OK rather than 0x0B to them	firmware volunteer	T11

#	Item	Consequence	Owner	Gate
5	Descriptor macro names ... are unverified against esp_tinyusb ~1.4.2 CLOSED 2026-09-02. They resolved against esp_tinyusb 1.4.5, and the _Static_assert on the MS OS 2.0 descriptor length (main.c:705) was evaluated and held	The descriptor set assembles at the asserted length. Whether Windows binds WinUSB to it is a different question and is T5	firmware volunteer	T5 on Windows 11
6	USB PID is a placeholder and the WinUSB GUID is the TinyUSB example GUID	Blocks the fleet, not the two prototypes. Browser authorisation is keyed to VID, PID and serial	programme	pid.codes allocation
7	sdkconfig.defaults is still missing five lines CLOSED 2026-09-02 (section 2.4). All five are in the file, and one of them was named wrongly here: the key is CONFIG_TINYUSB_VENDOR_COUNT, not CONFIG_TINYUSB_VENDOR_ENABLED, which does not exist and which ESP-IDF accepted in silence	The vendor class is enabled and the WebUSB interface exists in the built image. One half is still open: CONFIG_ESP_TASK_WDT_EN=y is set but nothing subscribes sample_task or the SD path to the watchdog, so a stalled SD write still does not reset the unit	firmware volunteer	T5; the watchdog subscription before T14

#	Item	Consequence	Owner	Gate
8	F-06 has been relaxed by ECO-EEG-025 to 90 s of ring plus microSD backfill; the 6 MiB ring gives 126 s as the simulator counts it, 124 s over the framed stream, and meets it either way (section 5.8)	Carried by RFQ-EEG-001 Rev E, DSN-EEG-003 Rev C and TST-EEG-004 Rev C. RING_BYTES is now 6 MiB in <code>main.c</code> and FW-D13 is closed. The residual work is the retransmit index of FW-D06 , which does not exist: CMD_RETRANSMIT still ignores its range and drains the ring destructively, so the ring is allocated but the backfill the 90 s depends on cannot yet be delivered	firmware volunteer	T15
9	<code>verify_stream.py</code> has no --session-id and roots the signature chain at 32 zero bytes; TST-EEG-004 T16 says the chain is rooted at the session identifier	Until the option is added, T16 must be run with an all-zero root and the deviation recorded	programme	before T16 is first executed
10	<code>verify_stream.py</code> reports pass when there are no SIGNATURE frames at all , and has no --compare for T14's card-versus-host check	The operator must assert <code>signature_blocks > 0</code> and diff the two reports by hand	programme	before T14 and T16

#	Item	Consequence	Owner	Gate
11	The microSD file format is now defined in full in section 5.9 and this document owns it; SVC-EEG-013 section 2 R2 cites that section and its /SESSIONS/...eegs layout with a 512-byte header is withdrawn. What is still open is the code. <i>Corrected 2026-09-02: this row said <code>sd_append()</code> and <code>sd_free_mb()</code> “are stubs”; they are written (<code>drivers.c:163</code> and <code>:175</code>) and have never addressed a card.</i> Nothing writes the sidecar, and <code>verify_stream.py</code> neither reads a sidecar nor checks the file hash	The definition no longer differs between documents, but no card has ever been written, so the layout is unexercised and T14 cannot run	firmware volunteer	T14
12	The host tool names diverge across the package: this document ships <code>tools/provision.py</code> and <code>tools/verify_stream.py</code>; other documents name <code>eegtest</code>, <code>prov_eeg.py</code> and <code>tools/eegtest/ota.py</code> for the same two programs	A test step that names a binary that does not exist cannot be executed as written	programme	before the first production run

#	Item	Consequence	Owner	Gate
13	The 6.81 MB FAT partition holds about 71 s of audio at 48 kHz 16-bit mono (calculated). Longer block audio must come from the card	F-11's pre-load model needs a decision	programme	Phase 2
14	Secure-boot v2 bootloaders are larger than plain ones; the partition table sits at 0x8000 , leaving 32 kB	If the signed bootloader overflows, CONFIG_PARTITION_TABLE_OFFSET must move and every offset in section 9 changes. Phase 1 does not hit this, because Phase 1 builds no signed bootloader	firmware volunteer	first secure build, Phase 2
15	The E-29 acoustic clamp is not implemented. SET_HP_LEVEL must clamp the codec volume register at the calibrated value; the codec driver is a stub	Calculated full-scale output is about 110 dB SPL against E-29's 100 dB SPL limit, so the requirement is not met until both the driver and the clamp exist	firmware volunteer	T17 and the E-29 type test
16	No safety engineer has reviewed this design, and no hardware has been built. <i>Corrected 2026-09-02: this row said "nothing in this package has been built". The firmware has been built and run under an emulator; no board has been fabricated and no unit exists</i>	Blocks use on a person. Does not block a build, a quote or firmware bring-up	programme	RISK-EEG-011

#	Item	Consequence	Owner	Gate
17	Static IRAM is 16,383 of 16,384 bytes used – one byte free (section 9). Turning off the unused SPI-slave and gptimer ISRs did not move the figure by a single byte, so it is not those	The next function marked IRAM_ATTR fails the link with an error naming a section and not a cause, and E-13's tone scheduler and E-12's onset detector are both still to be written. Either the pool is genuinely full and someone must choose what leaves IRAM, or esp_idf_size is reporting against a fixed 16 kB window that is not the real limit on an ESP32-S3 with octal SPIRAM and XIP. Deciding which needs the linker map read against hardware; it is not guessed at from here	firmware volunteer	the linker map read on a unit, before the next IRAM function is added. Also SIM-EEG-018 open item 1
18	CMD_ENTER_PROV answers 0x02 for “already provisioned”, where section 6.2 assigns 0x02 to bad length and 0x09 to locked – and 0x4B, in the same handler, answers 0x09 for the same condition (section 6.2)	An operator reading the status table alone hunts a framing fault that is not there. provision.py hints both readings at step 1, which is a workaround, not a fix. handle_provision() also returns 0x10-0x18 and 0x20 codes that section 6.2 does not define at all	firmware volunteer (main.c is not this document's to edit)	one line in main.c, before T6 is first executed

#	Item	Consequence	Owner	Gate
19	The abort-on-no-PSRAM is right for a bench and wrong for a field unit (section 8.2). A unit that aborts in a participant's home cannot be told apart from a dead battery or a bad cable, because it never enumerates and the browser tool cannot ask it what is wrong	The FW-D13 guard is a real improvement over asserting inside xRingbufferSend, and it is still the wrong failure mode in the field. What a field unit should do – enumerate, report the fault in STATUS, refuse to record – is not written	firmware volunteer	before the first unit leaves the programme
20	provision.py and README_provisioning.md still describe the old NVS layout – namespace "tioV" in the default nvs partition, with an unconditional erase – which drivers.c no longer does (sections 2.5 and 7.3). Their 0x4A/0x4B skip hints are also transposed: the config-write step reports a missing 0x4A and the calibration-read step a missing 0x4B	An operator following the script's own header would look for the constants in the wrong partition, and a skipped step would name the wrong opcode	firmware volunteer (those files are not this document's to edit)	before the first production run

Gap closure

This document closes the following entries from the v1 audit: fw-project-skeleton-missing (section 2), fw-source-defect-register (section 1.3), fw-sdkconfig-defaults (2.4 and 5.8), fw-partition-table (2.5), fw-pinmap-gpio35-37-conflict (section 3), fw-driver-stubs (1.2, with each stub bound to a test step), icd-command-and-frame-payloads (sections 5 and 6), usb-identity-and-vid-pid (4.3), provisioning-script-f18 (section 7), secure-boot-key-custody (7.5), firmware-release-package (section 9), eol-flashing-route (2.3, 7.1 and 9), host-production-test-tool (section 8), and the microSD file layout, which section 5.9 now defines for the whole package. It leaves open, and says so above: calibration-record-schema (CAL-EEG-012 owns the schema; section 7.4 fixes only the identity block), the host tool naming, and the block-audio storage decision.